The Capability–Reliability Gap in Artificial Intelligence: An Evidence-Graded Review, Maturity Framework, and Research Agenda, 2020–2026

Author: Angelis Pseftis
Affiliation: To be confirmed by the author before submission
Corresponding author: Contact details to be confirmed before submission
Intended venue: *Harvard Data Science Review*; article type and invitation status require editor confirmation
Review cutoff: August 28, 2026

Proposed CRediT author statement—author confirmation required

Subject to confirmation by Angelis Pseftis: Conceptualization, methodology, investigation, evidence curation, formal analysis, visualization, writing—original draft, writing—review and editing, and project administration. Final author approval of the contribution statement, evidence classifications, disclosures, and conclusions remains required before any journal submission.

Abstract

From 2020 through August 2026, artificial intelligence systems broadened from predominantly bounded prediction and generation tasks toward multimodal, tool-using configurations that can address difficult work in language, software, science, perception, and selected forms of control. Public evidence for operational dependability often remains less mature than evidence for selected task capability. This evidence-graded integrative review examines that cross-sectional asymmetry. It separates evidence strength from operational transfer and distinguishes demonstrated, repeatable, operational, and assured capability.

The reviewed evidence supports three bounded conclusions. First, capability gains are most defensible where objectives and environments are constrained and outputs are externally verifiable. Second, benchmark performance does not establish dependable operation under distribution shift, long horizons, ambiguous objectives, privileged tools, rare hazards, or consequential human reliance. Third, limited public evidence about tail behavior, correlated failure, and control effectiveness constrains justified high-consequence adoption even when average-case performance is useful.

The paper contributes an author-developed capability–assurance maturity framework, benchmark-validity criteria, an evidence-graded domain assessment, and a prioritized research agenda. Priorities include high-reliability long-horizon execution, runtime verification, dynamic and field evaluation, agent identity and authorization, containment and recovery, calibrated uncertainty, and incident learning. Two-, five-, and ten-year propositions are specified through prerequisites, indicators, and falsification criteria rather than an artificial-general-intelligence date. The synthesis supports treating frontier AI as a capable but partially characterized component whose permissible authority and consequence-specific evidence must be established for the actual system and use.

Keywords: artificial intelligence; AI evaluation; foundation models; operational reliability; AI assurance; technology forecasting

Media summary

Artificial intelligence systems can now write and analyze complex text, generate and repair software, interpret images and audio, operate digital tools, and contribute to selected scientific and robotic tasks. Those advances are real. They do not, by themselves, establish that an AI system can be trusted to perform a long, consequential workflow without unacceptable errors.

The difference matters because most public comparisons measure whether a model can produce a correct answer under a benchmark protocol. Operational users need a different body of evidence. They need to know how often the system fails, how failures cluster, whether performance survives changes in data and environment, whether the system

recognizes uncertainty, how it behaves when tools or permissions are available, whether an independent evaluator can reproduce the result, and whether harmful actions can be detected, reversed, and attributed.

This study calls that difference the capability–reliability gap. It separates four evidence states that are often collapsed into one claim: a system may demonstrate a behavior; repeat it under controlled conditions; deliver useful results in a bounded operational workflow; or support reliance through evidence and controls appropriate to the consequence of failure. Within the public evidence reviewed through August 28, 2026, some narrow systems had substantial operational evidence in defined domains. The review did not identify public evidence that any general-purpose foundation-model system satisfied a consequence-specific high-assurance claim across open-ended tasks. That is a bounded review finding, not proof of universal absence.

The study reviews evidence from language models, coding systems, agents, multimodal models, speech, scientific AI, robotics, autonomous driving, security, infrastructure, and governance. Across the included evidence, the more transferable results generally involved bounded objectives, measurable outputs, external verification, constrained authority, monitoring, and mature surrounding engineering. The evidence also motivates testing how failure changes with task duration, environmental novelty, model discretion, and permission rather than assuming that benchmark gains transfer unchanged.

The recommended research agenda therefore shifts attention from benchmark scores alone toward reliability curves, live and field evaluation, runtime verification, calibrated abstention, agent authorization, secure tool mediation, rollback, incident reporting, and system-level assurance cases. The outlook is presented as scenarios rather than a single prediction. Capability will probably continue to improve, but the social and economic effect of that improvement will depend on whether assurance science and engineering can keep pace.

1. Introduction

Artificial intelligence entered the 2020s with a well-established record in perception, recommendation, optimization, game playing, and other bounded tasks. During the following six years, large pretrained models and multimodal systems expanded the range of tasks that could be addressed through one model family. Systems learned from broad corpora, adapted through prompting or limited examples, generated executable code, invoked tools, interpreted combinations of text and images, and participated in increasingly long workflows. Scientific systems also produced results of independent significance in protein structure prediction, algorithm discovery, materials screening, and mathematical problem solving (Jumper et al., 2021; Fawzi et al., 2022; Merchant et al., 2023; Romera-Paredes et al., 2024; Trinh et al., 2024).

Two interpretations of this period are both inadequate. The first dismisses frontier systems as merely fluent autocomplete and therefore underestimates measurable advances in representation, planning, tool use, and economically useful task performance. The second extrapolates high benchmark scores or striking demonstrations into a claim of general, dependable autonomy. That interpretation ignores the dependence of observed performance on task formulation, prompt and scaffold design, inference budget, data exposure, external tools, evaluation harnesses, and the cost of failure. The empirical record supports a more precise conclusion: capability has broadened materially, while public assurance evidence remains less mature for many consequential, open-ended capability–context pairs. This is a cross-sectional evidence finding, not a measured comparison of progress rates.

This paper defines that asymmetry as the capability–reliability gap. Capability is the demonstrated ability of a system to produce a specified result under stated conditions. Reliability is the probability and distribution of acceptable performance across repeated use, including the effects of environmental change, interaction, duration, and failure recovery. Assurance is justified confidence, supported by evidence and controls, that the complete sociotechnical system will satisfy its requirements at a risk level appropriate to the consequences of failure. These are related but noninterchangeable properties. A model may be highly capable and poorly calibrated, useful with supervision and unsafe when granted irreversible authority, or reliable in a narrow operational design domain and unreliable outside it.

The distinction is not semantic. A static benchmark commonly estimates mean performance over a task sample. A production decision often depends on tail behavior, correlated failure, unauthorized action, recovery time, and the

validity of assumptions outside the test distribution. In a low-consequence drafting task, occasional errors may be inexpensive and readily corrected. In cyber defense, infrastructure control, clinical decision support, financial settlement, or military operations, the same error rate can be unacceptable because one failure may propagate, disclose protected information, or alter a physical or legal state. The appropriate standard therefore depends on the mission, authority, exposure, and reversibility of action rather than on a universal threshold for "intelligence."

The literature contains evidence of both progress and limitation. Scaling-law studies reported smooth loss relationships within observed model, data, and compute regimes; later work revised compute-optimal allocations and examined finite-data constraints (Kaplan et al., 2020; Hoffmann et al., 2022; Muennighoff et al., 2023). Benchmarks including MMLU, BIG-bench, GPQA, SWE-bench, MMMU, WebArena, and OSWorld documented capability on their stated knowledge, reasoning, coding, multimodal, and computer-use tasks (Hendrycks, Burns, et al., 2021; Srivastava et al., 2023; Rein et al., 2024; Jimenez et al., 2024; Yue et al., 2024; Zhou et al., 2024; Xie et al., 2024). Separate studies documented distribution-shift, calibration, long-context, confabulation, prompt-injection, specification-gaming, and goal-misgeneralization failure modes under specified experimental conditions; prevalence in any deployment must be measured separately (Koh et al., 2021; N. F. Liu et al., 2024; Farquhar et al., 2024; Greshake et al., 2023; Langosco et al., 2022; Hubinger et al., 2024).

Operational evidence reinforces the same conclusion. AI-assisted coding can accelerate constrained tasks, yet controlled studies have also found that experienced maintainers working in familiar, complex repositories may become slower even while believing they were faster (Peng et al., 2023; Becker et al., 2025). Autonomous driving demonstrates that machine-learned systems can operate at substantial scale within a defined operational design domain, but safety depends on sensors, maps, redundancy, simulation, remote support, incident response, and organizational controls rather than on a single model score. Legal and administrative failures involving fabricated citations show that a fluent model can be useful and still be an unacceptable authority when provenance is not enforced (Mata v. Avianca, Inc., 2023). These examples are not contradictions. They show that the unit of evaluation must be the deployed system and workflow, not the model in isolation.

This review makes five contributions. First, it provides an evidence-grading scheme that separates source strength from operational transferability. Second, it proposes four evidence states: demonstrated, repeatable, operational, and assured. Third, it proposes a two-axis maturity framework that rates capability maturity separately from assurance maturity. Fourth, it organizes research and engineering gaps into qualitative priority tiers using mission impact, cross-domain leverage, urgency, tractability, evidence deficit, and security consequence. Fifth, it develops conditional propositions for approximately two, five, and ten years, each with prerequisites, leading indicators, and falsification conditions.

The paper does not assign a date to artificial general intelligence. The reviewed communities do not use one operational definition consistently, and a single date would conceal the multidimensional nature of capability and reliability. The study instead asks what technical properties and evidence would be required to justify increasingly consequential forms of autonomy. This connects scientific progress to measurable engineering evidence without treating one benchmark as a universal intelligence scale.

2. Research Questions, Definitions, and Scope

The review addresses eight research questions.

What capabilities have been demonstrated across major AI domains since 2020, and what is the quality of the supporting evidence?

Which capabilities have progressed from demonstrations to repeatable or scaled operational use?

Where do benchmark results fail to predict operational performance, and why?

What technical, security, infrastructure, organizational, and governance factors produce the capability–reliability gap?

How should AI capability and assurance maturity be measured without reducing them to one scalar intelligence score?

Which research and engineering gaps have the greatest effect on safe and economically useful deployment?

What technical prerequisites would have to be satisfied for broader and longer-duration autonomy?

Which observable indicators would support or falsify plausible two-, five-, and ten-year scenarios?

For this paper, an AI system includes the trained model, inference configuration, prompts, retrieval components, memory, tools, orchestration logic, policy controls, monitoring, human roles, and operational environment that jointly produce behavior. A model is only one component of that system. A foundation model is a broadly trained model adapted to multiple downstream tasks, consistent with the research framing developed by Bommasani et al. (2021). An agent is a system in which a model selects or sequences actions that alter an external state, including actions executed through software tools, digital interfaces, or physical actuators. Autonomy refers to the extent to which those actions can be selected and executed without contemporaneous human authorization.

Capability is evaluated as a vector rather than a scalar. Relevant dimensions include task family, novelty, environmental variability, context length, tool availability, time horizon, acceptable latency, inference budget, output verifiability, authority, and consequence. The same model may be mature in one region of that vector and immature in another. A system that reliably translates routine text does not thereby acquire reliable causal reasoning, secure tool use, or physical dexterity.

Reliability means more than repeatability. It includes the frequency, severity, detectability, and recoverability of failure. For an AI-enabled system, reliability analysis must also consider correlated errors, hidden state, data drift, model updates, tool and retrieval failures, adversarial influence, and human reliance. Assurance is the structured argument, supported by evidence, that requirements and risk controls are adequate for the intended use. It includes verification, validation, security, safety, configuration control, monitoring, and incident learning.

The primary review window is January 2020 through August 28, 2026. Earlier work is included when it defines a concept or method required to interpret the period, including calibration, automation bias, safety engineering, dataset documentation, and the statistical demonstration of rare-event bounds. The scope includes foundation and multimodal models, reasoning, coding, agents, tool use, retrieval and memory, scientific AI, reinforcement learning, robotics, autonomous vehicles, evaluation, interpretability, security, compute, data, energy, standards, and governance. The paper does not attempt a complete catalog of every model release or benchmark score. Rapidly changing vendor leaderboards are treated as horizon-scanning evidence unless independently reproduced.

3. Methodology

3.1 Review design and source selection

This study is an evidence-graded integrative review. It is not represented as a completed exhaustive systematic review or a statistical meta-analysis. The literature spans incompatible outcome measures, rapidly changing model versions, private training data, nonstandard inference budgets, and benchmark families that are repeatedly replaced as they saturate or become contaminated. Pooling those values into one effect size or “frontier intelligence” index would violate comparability assumptions.

The evidence base contains peer-reviewed journal and conference papers, benchmark publications, official standards and standards catalogs, government technical reports, court and regulator records, independent evaluations, and model-developer technical documentation. The August 2026 audit used targeted searches of publisher and conference records, government and regulator sites, standards catalogs, Crossref and OpenAlex metadata, and primary project pages. Search terms combined foundation models, multimodality, agents, coding, tool use, robotics, scientific machine learning, evaluation, reliability, robustness, calibration, uncertainty, confabulation or hallucination, causality, interpretability, adversarial machine learning, prompt injection, data provenance, compute, energy, governance, and assurance. Backward and forward citation tracing connected quantitative claims to original methods and material limitations. The supplementary search log records the sources, queries, dates, and cutoff-update checks. Developer documentation is retained only for bounded developer-reported claims and is not treated as independent validation.

Sources were included when they provided at least one of the following: a defined task and evaluation method; empirical results with enough detail to interpret conditions; a reproducible benchmark or dataset; field or deployment

evidence; a formal standard or risk framework; a documented incident; or a technical analysis of a material constraint. Sources were excluded from core conclusions when they relied only on marketing language, did not identify the evaluated system or conditions, treated a single anecdote as a population estimate, or repeated a quantitative claim without a traceable primary source.

Because the evidence frontier changes faster than traditional publication cycles, publication status and current relevance are evaluated separately. A peer-reviewed result may no longer describe a frontier system, while a current system card may be timely but interested-party evidence. The audit therefore classifies each material statement as a sourced empirical or normative claim, a derived result, an author-developed framework, an analytical judgment, a recommendation, or a conditional scenario. E0–E4 evidence and T0–T4 transfer grades apply to sourced empirical and operational claims; nonempirical entries are marked not applicable and retain their stated basis.

3.2 Evidence-grading model

Table 1 defines the evidence-strength scale. The grade attaches to a claim, not permanently to a source. A peer-reviewed benchmark paper may provide strong evidence that a test was constructed as described but weaker evidence that the test predicts field performance. A developer system card may provide direct evidence of the developer's stated procedure but only preliminary evidence of comparative capability.

Table 1. Evidence-strength scale

Grade	Description	Typical evidence	Permitted interpretation
E0	Unsupported or nontraceable	Marketing assertion, unattributed number, unverifiable anecdote	Excluded from substantive conclusions
E1	Preliminary or interested-party evidence	Developer-reported evaluation, preprint without independent replication, single case	Evidence that a result was reported under stated conditions
E2	Credible primary or single-source evidence	Peer-reviewed study, independent evaluation, official record, reproducible benchmark paper, or auditable derivation	Supports a bounded claim with explicit limitations
E3	Convergent independent evidence	Multiple independent studies, repeated benchmark families, cross-method agreement	Supports a general pattern within defined conditions
E4	Sustained operational evidence	Scaled field data, repeated outcomes, mature incident and change controls	Supports operational claims within the documented design domain

Operational transferability is graded separately.

Table 2. Operational-transfer scale

Grade	Evaluation context	Transfer implication
T0	Toy, synthetic, or poorly specified task	Little direct evidence about real use
T1	Static benchmark or controlled laboratory task	Evidence of task competence under the test distribution
T2	Realistic controlled workflow or field pilot	Moderate evidence for a bounded operational process
T3	Scaled operation in a defined domain	Strong evidence for that domain, not for unrestricted use
T4	High-consequence operation with audited assurance evidence	Evidence appropriate to a specified assurance claim

A claim can therefore be coded E3/T1, meaning that independent studies consistently reproduce a benchmark phenomenon whose operational relevance remains limited, or E2/T3, meaning that a credible field study provides direct operational evidence but has not yet been replicated across organizations. Transfer is marked not applicable for legal status, formal derivations, author proposals, and recommendations. This prevents source prestige, mathematical validity, or normative authority from being confused with ecological validity.

3.3 Benchmark-validity assessment

Benchmarks were assessed against nine questions: whether the construct is clearly defined; whether the test distribution represents the claimed use; whether training contamination is plausible; whether the benchmark is saturated; whether scoring is sensitive to prompts or harness design; whether tools and inference budgets are reported; whether results are reproducible across model versions; whether uncertainty is quantified; and whether benchmark performance has been linked to field outcomes.

Static public benchmarks remain useful for comparison, diagnosis, and scientific progress. They are not neutral measuring instruments once they become optimization targets. Data may enter training corpora, developers may tune prompts and scaffolds to the test, and benchmark-specific techniques may improve scores without producing equivalent generalization. Rephrased-sample studies show that contamination analysis itself is sensitive to how overlap is defined, while live and time-separated benchmarks reduce—but do not eliminate—exposure risk (Yang et al., 2024; White et al., 2025; Zhang et al., 2025). Dynamic evaluation, private test sets, adversarial testing, and field trials reduce some problems but introduce cost, access, governance, and reproducibility tradeoffs (Kiela et al., 2021; Liang et al., 2023; National Institute of Standards and Technology, 2026d).

3.4 Capability and assurance maturity

The maturity framework uses two axes. Capability maturity asks whether the system can perform the task across increasingly realistic conditions. Assurance maturity asks whether the evidence and controls justify reliance at the proposed consequence level. The axes are separated because high capability does not imply mature assurance, and a highly assured system may intentionally support a narrow capability.

Ratings are assigned to a defined capability–context pair rather than to a model brand. For example, “speech transcription in a known language and acoustic environment” is a ratable function; “general intelligence” is not. The assessment considers task validity, external replication, distribution shift, calibration, security, monitoring, human workload, failure recovery, configuration control, and incident evidence.

3.5 Gap-prioritization method

Research and engineering gaps are organized into qualitative priority tiers using six author-defined considerations: mission and societal impact, cross-domain dependency leverage, security and safety consequence, urgency, evidence deficit, and tractability over the next five years. The package does not contain independently elicited criterion scores or a valid sensitivity analysis. The revised agenda therefore removes decimal rankings and treats the tiers as transparent analytical judgments for portfolio discussion, not measurements.

Technical difficulty is reported separately. A difficult problem may still receive the highest priority because it blocks many consequential uses; a tractable problem may sit in a lower tier if its effect is narrow. Items within a tier are not ordered, and reasonable decision-makers may assign different tiers under different missions or risk tolerances.

3.6 Scenario method

The outlook uses conditional scenarios rather than point predictions. Each scenario identifies the technical prerequisites, deployment assumptions, leading indicators, and falsification conditions that would have to be observed. The horizons are approximately two years (2028), five years (2031), and ten years (2036) from the review cutoff. Confidence declines with horizon. Scenarios describe patterns of capability and assurance, not a date for AGI.

3.7 Methodological limitations

The review has material limitations. It is a targeted, evidence-graded integrative review, not an exhaustive systematic review or meta-analysis; database coverage, screening decisions, and negative searches therefore do not prove universal absence. Training data, evaluation prompts, inference budgets, and post-training methods are often proprietary. Current capability evidence is frequently developer-reported before independent evaluation, publication and benchmark selection favor successful results, model and test replacement weakens longitudinal comparison, and operational incident data remain incomplete. The maturity ratings and priority tiers are unvalidated author-developed instruments without inter-rater reliability evidence. No proprietary deployment evidence, external hostile review, or independent replication was available for this audit. These constraints limit what can be concluded about deployed systems.

4. Evolution of the Frontier, 2020–2026

4.1 Scaling, transfer, and learned models of environments

The period began with two research lines that shaped the subsequent frontier. Neural scaling-law work showed that language-model loss followed relatively smooth power-law relationships with model size, data, and compute across the observed regimes (Kaplan et al., 2020). This provided an empirical basis for investing in larger training runs. The result was not that scale guaranteed any particular capability, but that aggregate predictive performance could be forecast more reliably than many researchers had expected.

At the same time, MuZero demonstrated that a reinforcement-learning system could learn a model sufficient for planning without being given the rules or transition dynamics of Go, chess, shogi, or Atari in an explicit form (Schrittwieser et al., 2020). The learned representation was not a general world model, yet the result showed that planning-relevant latent dynamics could support strong control in closed environments. Together, scaling and learned planning encouraged a shift from hand-engineered task pipelines toward broadly trained systems whose downstream abilities emerged from representation learning, post-training, and inference.

The limits were visible from the beginning. Scaling curves described average loss within a family and compute regime, not robustness, truthfulness, causal understanding, or safe action. Closed games provided complete rules, resettable environments, abundant simulation, and unambiguous rewards. Those properties are absent from most institutional and physical operations.

4.2 Foundation models, instruction following, and compute-optimal training

By 2021 and 2022, the foundation-model paradigm had become a useful description of systems trained broadly and adapted across tasks (Bommasani et al., 2021). Instruction tuning and reinforcement learning from human feedback improved the usability of language models by aligning outputs more closely with user intent and conversational norms (Ouyang et al., 2022). These methods materially changed interaction quality, but they optimized observable behavior under sampled preferences. They did not provide a proof that internal objectives were stable under distribution shift or that a model would remain safe when connected to tools.

Compute-optimal scaling work then showed that many large models had been undertrained relative to their parameter count. Hoffmann et al. (2022) found that, under their assumptions, allocating more training tokens to a smaller model could outperform a much larger model trained on fewer tokens. Later work examined data-constrained regimes, repeated data, and the interaction between training and inference costs (Muennighoff et al., 2023; Sardana et al., 2024). The strategic implication was that “more parameters” was never a sufficient description of progress. Data quality, token allocation, architecture, post-training, retrieval, and inference-time computation all affected capability and cost.

4.3 Multimodality, tools, and scientific systems

From 2022 through 2025, models increasingly integrated language with images, audio, code, tools, and action. ReAct interleaved reasoning traces with external actions, Toolformer explored self-supervised tool use, and agent benchmarks began testing navigation through websites and computer interfaces (Yao et al., 2023; Schick et al., 2023; Zhou et al., 2024; Xie et al., 2024). Multimodal benchmarks such as MMMU tested expert-level questions containing diagrams, charts, and images, while Video-MME extended evaluation across temporal media (Yue et al., 2024; Fu et al., 2025).

Scientific AI produced some of the clearest evidence that machine learning could contribute beyond text imitation. AlphaFold2 achieved a major advance in protein structure prediction and supported a large public structure database (Jumper et al., 2021; Varadi et al., 2022). AlphaTensor found improved matrix-multiplication algorithms for selected cases, FunSearch combined language-model proposals with automated evaluation to discover new mathematical and algorithmic results, and AlphaGeometry solved difficult olympiad geometry problems through a neuro-symbolic system (Fawzi et al., 2022; Romera-Paredes et al., 2024; Trinh et al., 2024). These systems succeeded because the domains provided strong structure, evaluators, simulators, or formal checks. Their achievements support the value of AI in science, but they do not imply that unconstrained conversational models can autonomously conduct reliable science.

4.4 Agents, live evaluation, and the return of systems engineering

By 2024 and 2025, evaluation increasingly shifted from isolated answers toward repository repair, browser use, terminal interaction, and longer tasks. SWE-bench used real software issues and repositories, WebArena supplied realistic websites, and OSWorld evaluated open-ended tasks across computer environments (Jimenez et al., 2024; Zhou et al., 2024; Xie et al., 2024). METR and related work modeled task success as a function of the time required by a human expert (Kinniment et al., 2023; Kwa et al., 2025). METR's January 2026 Time Horizon 1.1 revision used 228 software-oriented tasks, including 31 estimated at eight hours or longer, and reported substantial sensitivity to task-suite composition; only five of those 31 long-task baselines were measured rather than estimated (METR, 2026b). These are suite- and scaffold-specific estimates of task difficulty, not universal measures of agent wall-clock duration.

This shift exposed a systems problem. Agent performance depended on the model, prompt, memory, tool interface, environmental state, retry policy, verifier, and resource budget. A leaderboard entry no longer described a self-contained algorithm. It described a configured system. The same transition occurred in retrieval-augmented generation and multimodal applications. As the model became one component in a larger control loop, software assurance, cybersecurity, identity, observability, and human factors became first-order determinants of outcome.

The 2026 frontier is best understood in that context. Provider-reported scores changed rapidly, while independent evaluation and operational evidence remained uneven. NIST launched an AI Agent Standards Initiative in February 2026, and the National Cybersecurity Center of Excellence issued a draft concept paper on software- and AI-agent identity and authorization; these records show that interoperability, security, identity, and authorization were active standardization topics, not that final agent standards or validated controls already existed (National Cybersecurity Center of Excellence, 2026; National Institute of Standards and Technology, 2026a). The decision-relevant change is the emergence of partially autonomous systems whose actions can accumulate over time, increasing both potential utility and the consequence of uncharacterized failure.

5. Current Capability Frontier

5.1 Language, knowledge, and bounded reasoning

Large language models now perform a broad range of linguistic and knowledge tasks at levels that make them useful for drafting, classification, extraction, tutoring, summarization, translation, question answering, and analytical assistance. MMLU documented a substantial step beyond earlier narrow language evaluations through 57 tasks spanning academic and professional subject areas (Hendrycks, Burns, et al., 2021). BIG-bench expanded the task distribution and demonstrated both smooth and discontinuous-looking performance changes across scale (Srivastava et al., 2023). GPQA raised the difficulty with questions designed to challenge nonexpert humans and resist simple search (Rein et al., 2024). Successor tests, including MMLU-Pro, LiveBench, and newly collected or time-separated evaluations, have been introduced because public static benchmarks can saturate, leak into training data, or cease to separate frontier systems (Wang et al., 2024; White et al., 2025; Zhang et al., 2025).

These results establish broad learned competence. They do not establish a unitary reasoning faculty. Performance varies with wording, decomposition, sampled inference, context, tool access, and whether correctness can be checked. A system may answer graduate-level science questions while failing a simpler problem that requires tracking constraints across a long interaction. Apparent step changes may also reflect metric choice. Schaeffer et al. (2023) showed that some reported "emergent abilities" could be produced by nonlinear or discontinuous evaluation metrics even when underlying model performance changed smoothly. The practical lesson is not that emergence is unreal, but that claims about qualitative phase transitions require measurement designs that rule out artifacts.

Reasoning performance is strongest when a problem can be reduced to steps with external verification. Code execution, calculators, search, theorem provers, and sampling followed by ranking can all improve outcomes. Inference-time computation therefore changes the capability surface. It also changes cost, latency, attack surface, and the number of internal decisions that may fail. The correct comparison is not merely answer accuracy before and after "more thinking." It is verified task utility per unit of compute, time, and autonomous authority.

Truthfulness remains a separate property. TruthfulQA demonstrated that models could reproduce common misconceptions rather than provide truthful answers (Lin et al., 2022). FActScore and related work decomposed long-form generation into atomic claims and found that factual precision varied materially across systems and settings (Min et al., 2023). SelfCheckGPT showed that sampling consistency could help detect unsupported statements without access to model internals, while semantic-entropy methods used variation in meaning across outputs to identify uncertainty (Manakul et al., 2023; Farquhar et al., 2024). These methods reduce risk but do not convert generation into an authoritative database. They are detectors with their own false-positive and false-negative rates.

The mature operational pattern is therefore assisted language work with provenance, retrieval, structured output, and accountable review. Unsupervised generation remains inappropriate where a fabricated source, omitted exception, or subtly wrong causal claim can directly create legal, clinical, financial, or safety consequences. The distinction is visible in legal hallucination incidents: the model's language capability was adequate to produce plausible filings, but the workflow lacked a control that required verification against authoritative records (Mata v. Avianca, Inc., 2023).

5.2 Software engineering and digital agents

Software is unusually favorable for AI because many outputs can be executed, tested, statically analyzed, compared against specifications, or reviewed through version control. Code also provides abundant training data and a language in which small improvements are economically valuable. Competition-level programming systems and code-oriented foundation models showed that generated programs could solve increasingly difficult problems (Li et al., 2022). SWE-bench then shifted evaluation toward real repository issues, exposing the need to understand codebases, modify multiple files, and satisfy tests (Jimenez et al., 2024).

The original SWE-bench paper reported 1.96 percent issue resolution for Claude 2, the strongest of its reported baselines, on a 2,294-task dataset drawn from 12 Python repositories (Jimenez et al., 2024). Later configured systems reported much higher scores, aided by stronger models, repository tools, refined scaffolds, and curated benchmark variants. The direction of benchmark improvement is clear, but exact longitudinal comparisons are not. Different benchmark versions remove problematic tasks, use different test harnesses, allow different tools, and consume different inference budgets. A result on a curated "verified" subset is not directly equivalent to a result on the original set, and a patch that passes automated tests is not necessarily acceptable to a maintainer. Provider audits in 2026 reported contamination and test-design defects in SWE-bench Verified and material signal-quality problems in SWE-Bench Pro; those interested-party audits support retiring simple cross-version score comparisons, not substituting a provider claim for independent measurement (OpenAI, 2026a, 2026b).

Productivity studies illustrate the transfer problem. Peng et al. (2023) reported that participants using GitHub Copilot completed one bounded JavaScript HTTP-server task 55.8 percent faster on average. Becker et al. (2025), in a randomized study of 16 experienced open-source developers completing 246 issues in repositories they knew, estimated that early-2025 tools increased completion time by 19 percent (95 percent confidence interval: 2–39 percent slower) even though participants expected a 24 percent speedup and perceived a 20 percent speedup. METR subsequently changed the follow-up experiment design in February 2026; that design note is prospective and does not revise the published early-2025 estimate (Becker et al., 2026). The studies measured different populations, codebases, tasks, and tool generations. Together they show that productivity effects can differ sharply by task and population; neither estimate supports a universal productivity effect.

Security evidence adds another qualification. Studies of code assistants have identified insecure suggestions and cases in which users produced more vulnerable code when relying on AI assistance (Pearce et al., 2022; Perry et al., 2023). The implication is not to reject coding assistants. It is to place them inside a secure development lifecycle with tests, static and dynamic analysis, dependency controls, code review, provenance, and clear ownership. The model should not be the final acceptance authority for code it generated.

Digital agents generalize the coding pattern to browsers, terminals, business applications, and APIs. ReAct, Reflexion, WebArena, AgentBench, and OSWorld show that configured systems can plan, call tools, observe results, and revise actions (Yao et al., 2023; Shinn et al., 2023; X. Liu et al., 2024; Zhou et al., 2024; Xie et al., 2024). OSWorld 2.0, released as a June 2026 preprint, introduced 108 longer workflows and reported 20.6 percent strict completion for its

best evaluated configuration under a 500-step protocol (Yuan et al., 2026). That protocol is not directly comparable to OSWorld 1.0. Across these studies, long, stateful workflows remained substantially harder than short question-answering tasks. Reported failures included losing state, misreading interfaces, repeating ineffective actions, selecting a locally plausible but globally wrong step, and continuing after the environment had diverged from the assumed plan.

Agentic capability should therefore be measured at multiple reliability levels. A 50 percent success horizon answers how long a task can be before a system succeeds half the time. It does not answer how long the system can operate with 99.9 percent reliability, low side effects, and bounded recovery cost. For most consequential uses, the latter is the relevant question. Current evidence is too sparse to estimate extreme reliability across diverse long tasks with confidence (Kinniment et al., 2023; Kwa et al., 2025).

5.3 Vision, multimodality, speech, and temporal understanding

Vision systems reached high benchmark performance before the current foundation-model period, yet natural distribution shifts continued to expose brittleness. ImageNet-A collected naturally occurring examples that strong ImageNet classifiers frequently misclassified, and WILDS documented performance degradation across hospitals, cameras, geography, time, and other real shifts (Hendrycks, Zhao, et al., 2021; Koh et al., 2021). Shortcut learning provides a general explanation: a model can exploit predictive features that are stable in the training data but unrelated to the intended causal structure (Geirhos et al., 2020).

Multimodal foundation models broadened the set of visual tasks by combining images with language. MMMU evaluated questions across disciplines that required interpreting figures, diagrams, charts, and other visual evidence (Yue et al., 2024). MM-Vet and related benchmarks tested combinations of recognition, knowledge, spatial reasoning, and generation (Yu et al., 2024). Video-MME extended evaluation to videos ranging from 11 seconds to one hour and included visual, subtitle, and audio inputs (Fu et al., 2025). These are benchmark results; they do not by themselves establish dependable object persistence, temporal ordering, fine-detail grounding, or causal attribution in a deployed workflow.

Speech recognition and translation are comparatively mature. Whisper demonstrated robust multilingual speech recognition from large-scale weak supervision, and SeamlessM4T integrated multiple speech and text translation tasks across many languages (Radford et al., 2023; Seamless Communication et al., 2023). Production systems can deliver valuable transcription and translation at scale. Reliability remains sensitive to accents, noise, domain terminology, code switching, speaker attribution, low-resource languages, and hallucinated transcript segments. Where transcripts become legal, intelligence, clinical, or safety records, confidence, audio retention, human correction, and provenance are required.

The central multimodal assurance problem is cross-modal consistency. A system may correctly recognize text in an image while inventing a relationship among objects, or transcribe speech accurately while assigning it to the wrong speaker. Evaluation must therefore separate perception, grounding, inference, and generation. A single end-to-end score makes it difficult to diagnose which component failed and whether a downstream control could have caught the error.

5.4 Scientific AI and verifiable discovery

Scientific AI provides direct counterexamples to the claim that machine learning can produce only fluent imitation. AlphaFold2 greatly outperformed competing methods in the CASP14 blind benchmark, and the AlphaFold Protein Structure Database subsequently provided more than 214 million predicted structures with confidence metrics (Jumper et al., 2021; Varadi et al., 2022; Barrio-Hernandez et al., 2023). AlphaTensor found provably correct algorithms for selected matrix-multiplication cases; GNoME reported 2.2 million structures stable relative to the Materials Project, including 381,000 on an updated convex hull and 736 matches to independently reported experimental structures; FunSearch found new cap-set constructions and online bin-packing heuristics; and AlphaGeometry solved 25 of 30 translated olympiad-geometry problems within its formalized domain (Fawzi et al., 2022; Merchant et al., 2023; Romera-Paredes et al., 2024; Trinh et al., 2024).

These cases show that machine learning can produce artifacts assessed through structural benchmarks, exact computation, formal proof, density-functional calculations, database matching, or experiment rather than subjective

impressions of fluency. They also show that verifier quality is system-specific: a formal proof, compiler, physical calculation, database match, and experiment have different error, independence, and transfer properties. A strong verifier for one bounded artifact does not confer assurance on an end-to-end scientific workflow.

The results do not justify a general claim that a language model can independently formulate, execute, interpret, and reproduce open-ended science. Scientific workflows contain hidden confounders, instrument error, literature bias, tacit knowledge, changing hypotheses, and ethical constraints. Laboratory autonomy requires safe physical control and traceable data. The near-term opportunity is a partnership in which AI expands search, proposes hypotheses, writes analysis code, and connects evidence while formal methods and accountable experts validate outputs.

5.5 Robotics and embodied autonomy

Embodied AI combines perception, language, planning, control, and physical interaction. PaLM-E reported positive transfer across controlled embodied reasoning and manipulation tasks; RT-2's developer evaluation reported improved semantic generalization in controlled robot trials while retaining physical-skill limits from the robot-data distribution; and Open X-Embodiment pooled more than one million real-robot trajectories from 22 embodiments and more than 500 skills for controlled multi-laboratory evaluation (Driess et al., 2023; Brohan et al., 2023; O'Neill et al., 2024). DreamerV3 used fixed hyperparameters across more than 150 mostly simulated control tasks and learned to obtain a Minecraft diamond from scratch; that 2025 peer-reviewed result strengthens simulated-control evidence, not field-robot reliability (Hafner et al., 2025).

Physical reliability remains materially harder than digital generation. Contact dynamics, occlusion, wear, sensor failure, latency, human proximity, and irreversible action create failure modes that cannot be removed by sampling another answer. Demonstrations are often collected in structured laboratories with known objects and safety operators. Generalization to unfamiliar homes, disaster environments, or industrial edge cases remains limited.

Autonomous driving demonstrates both the potential and the boundary. For Waymo's rider-only Level 4 service in the studied locations and periods, a peer-reviewed developer analysis and an independent Insurance Institute for Highway Safety analysis reported lower crash-involvement rates than matched human benchmarks (Kusano et al., 2024; Teoh et al., 2026). Those results are specific to the named vehicles, operational design domains, outcome definitions, reporting adjustments, exposure denominators, and time windows; they do not establish safety in arbitrary geography, weather, road design, or policy conditions. Published safety cases for some Level 4 services describe redundant sensing, maps, simulation, fleet operations, remote assistance, maintenance, and incident processes, but component effectiveness remains system-specific. After the October 2, 2023 Cruise collision, the driverless vehicle dragged a pedestrian approximately 20 feet; NHTSA found incomplete incident reports, and Cruise later admitted submitting a false record intended to influence the federal investigation (National Highway Traffic Safety Administration, 2024; U.S. Attorney's Office, Northern District of California, 2024).

The transferable lesson is that embodied autonomy matures by constraining the domain, measuring exposure, collecting incidents, and engineering fail-safe behavior. "General robot intelligence" is not a substitute for an operational design domain.

5.6 Retrieval, context, memory, and adaptation

Retrieval-augmented generation improves access to current or private information by placing selected documents in the model's context. It can reduce unsupported generation and provide citations, but reliability depends on retrieval recall, document quality, chunking, ranking, access control, and whether the model uses evidence faithfully. Poisoned or malicious retrieved content can also become an instruction channel. Retrieval should therefore be treated as a security and data-provenance subsystem, not merely a factuality feature.

Long context windows increase the quantity of available information but do not guarantee that the model will use all parts effectively. "Lost in the middle" experiments found that performance could degrade when relevant information appeared in the middle of long contexts rather than at the beginning or end (N. F. Liu et al., 2024). LongBench and RULER further showed that nominal context length and effective task performance diverge across retrieval, reasoning, and aggregation tasks (Bai et al., 2024; Hsieh et al., 2024). Long context is therefore capacity, not memory assurance.

Persistent memory introduces additional concerns. Stored summaries may contain errors that become premises for later action. User profiles can create privacy risk. Model updates may change how old memories are interpreted. Continual learning can reduce staleness but can also cause catastrophic forgetting, policy drift, or adversarial persistence. Dependable memory requires provenance, access control, confidence, expiration, correction, and a distinction between authoritative state and model-generated recollection.

5.7 Evidence summary by capability domain

Table 3. Claim-bounded capability evidence and residual limitations

Capability claim	Strongest included evidence context	Evidence/ transfer	Author assessment of operational use	Principal residual limitation
Assistive language generation and transformation	Controlled studies and documented human-reviewed workflows	E3/T2	Useful when evidence and accountable review are part of the workflow	Factuality, provenance, calibration, and domain-specific exceptions
Knowledge and bounded reasoning	Static, live, and tool-assisted benchmarks	E3/T1	Useful on bounded tasks with reliable external checks	Task sensitivity, contamination, and weak transfer to long workflows
Coding assistance	Repository benchmarks and randomized productivity studies	E3/T2	Useful with tests, secure review, and configuration control	Repository complexity, maintainability, insecure suggestions, and variable productivity transfer
Digital agents	Realistic controlled browser, terminal, and application environments	E2/T2	Limited unattended use; system configuration is decisive	Long-horizon state, side effects, recovery, and authorization
Multimodal understanding	Expert and temporal benchmark suites	E3/T1	Useful with human confirmation in bounded workflows	Grounding, fine detail, temporal order, attribution, and distribution shift
Speech recognition and translation	Large benchmark suites and documented deployed use in defined conditions	E3/T2	Mature for selected assistive uses, not for every record context	Low-resource languages, noise, speaker attribution, and record integrity
AlphaFold2 prediction and AlphaFold Database operation	Blind structural benchmark and scaled public database	E3/T1 for prediction; E3/T3 for database operation	Strong for specified prediction and database use	Predicted structure is not experimental determination; confidence and use limits remain
Scientific candidate, algorithm, and formal-problem systems	Single-system formal, computational, or experimental evaluation	E2/T1	Promising and verifier-dependent for named tasks	Verifier error, experimental validity, workflow integration, and open-ended autonomy
Generalist vision-language-action robotics	Controlled laboratory and multi-laboratory research evaluations	E2/T1	Immature for unattended physical operation	Physical distribution shift, recovery, safety, and exposure evidence
Named Level 4 rider-only service in a defined operational design domain	Sustained field exposure, incident records, and matched crash studies	E4/T3 for operational exposure; E3/T3 for comparative crash evidence	Mature only for the named system, versions, locations, and periods studied	Domain expansion, rare hazards, denominator sensitivity, and governance consistency
Open-ended long-horizon autonomy	Agent benchmarks and limited controlled studies	E2/T1	Insufficient evidence for high-consequence unattended use	Tail reliability, side effects, security, and measurement of rare failure

The review groups these claims into three interpretive classes rather than validated taxonomic categories: claims with system-specific operational evidence; claims with credible controlled evidence but limited operational transfer; and preliminary or highly scaffold-dependent claims. The class assignment is traceable in the supplement and should not be generalized beyond the named system, task, configuration, and evidence context.

6. The Capability–Reliability Gap

6.1 Four states that should not be conflated

This review proposes four evidence states for a defined capability claim and use context. They are an analytical framework, not a validated consensus maturity standard.

A **demonstrated capability** has been observed under a specified setup. A demonstration may be scientifically important, but it can depend on selection, retries, expert prompting, or a narrow task distribution.

A **repeatable capability** recurs across trials under the same specified procedure and configuration. **Reproducibility** is a stronger and separate property: an independent team obtains a materially consistent result using the reported method and evidence. Neither property, by itself, establishes performance under operational variation.

An **operational capability** produces useful outcomes in a real workflow under defined constraints. The claim must identify its operating context, users, interfaces, environmental conditions, integration, latency, cost, and recovery; automated-driving systems commonly express such limits as an operational design domain.

An **assured capability** is supported by evidence and controls that justify reliance at a specified consequence level. A defensible claim normally addresses requirements, hazards and threats, verification and validation, configuration management, monitoring, incident response, and accountable treatment or acceptance of residual risk, tailored to applicable law, standards, and mission authority.

The gap arises when evidence from an earlier state is used to justify a later one. A benchmark demonstration is cited as proof of workforce productivity, a controlled agent task is cited as proof of autonomous business operation, or a refusal benchmark is cited as proof that a model cannot be induced to perform harmful actions. These inferences may eventually be true, but they require additional evidence. Figure 1 maps this distinction onto the proposed capability and assurance axes.

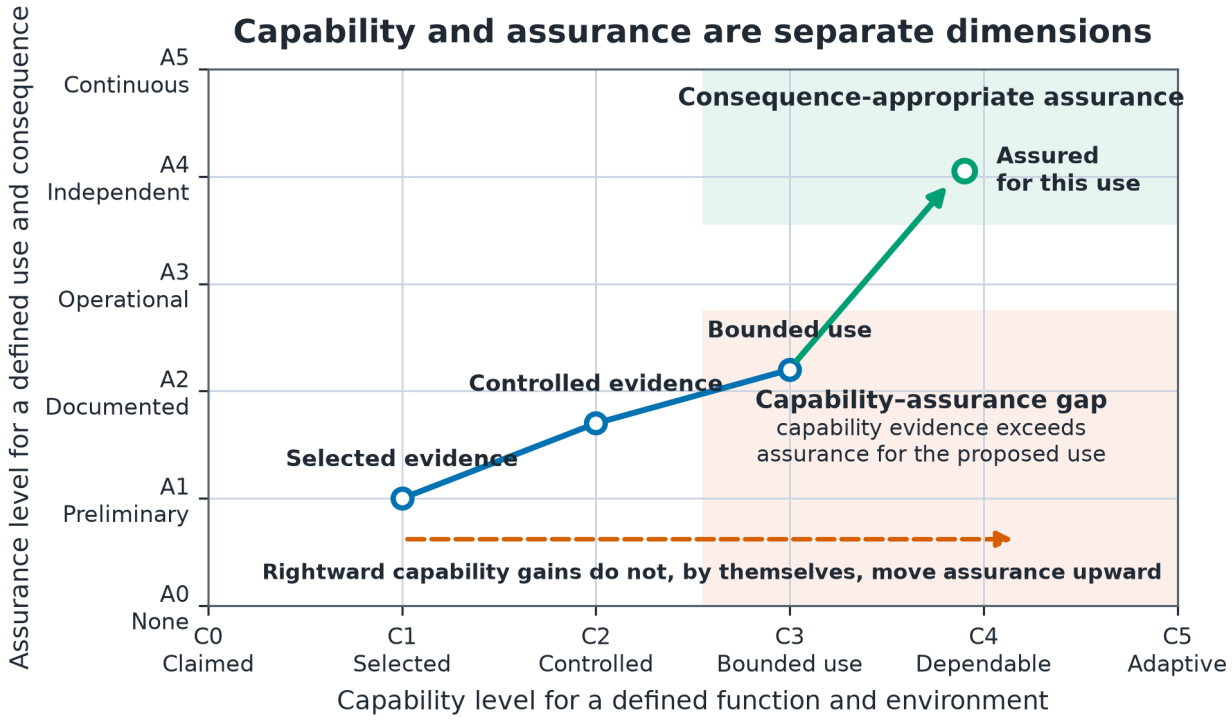


Figure 1. Capability level and assurance level are separate dimensions. The blue path illustrates movement from selected evidence (C1/A1) through controlled evidence (C2) to bounded operational use (C3), while the green path to an A4-level assurance claim requires independent, consequence-appropriate evidence and controls. The orange path shows that rightward capability gains do not, by themselves, move assurance upward. C0–C5 and A0–A5 are defined in Table 4. This is an author-developed conceptual framework, not empirical data or a validated measurement instrument.

6.2 Reliability compounds across long workflows

Long-horizon systems expose a mathematical problem that benchmark averages obscure. For a pure series workflow in which every one of n steps must succeed, the step-success events are independent, and every step has the same success probability p, the probability that every step succeeds is:

$P(\text{workflow success}) = p^n$.

At 99 percent per-step success, a 100-step pure series workflow succeeds without a failed step about 36.6 percent of the time. At 99.9 percent per-step success, the result is about 90.5 percent. Reaching 99 percent end-to-end success across 100 steps requires 99.98995 percent success at each step under the stated model. More generally, all-step success is the product of the conditional success probabilities. Common-cause errors, cascades, or declining conditional success can make the independence product optimistic; other dependence structures can move it upward. Dependence must therefore be modeled rather than assigned one universal direction.

Not every workflow requires an error-free series of unverified steps. Retry, verification, checkpointing, recovery, optional branches, and redundancy require different models and can improve system reliability when detection coverage, independence, retry side effects, and recovery success are measured. Long-horizon reliability is therefore a systems property, not an extrapolation of one model-accuracy estimate.

Rare-event bounds are statistically demanding. With zero failures in n independent, identically distributed, representative Bernoulli trials and a fixed event definition and configuration, the approximate one-sided 95 percent upper confidence bound on failure probability is $3/n$, the "rule of three" (Hanley & Lippman-Hand, 1983). Exact binomial calculations require at least 2,995 failure-free trials to bound failure probability below 10^{-3} and at least 299,572 to bound it below 10^{-5} at 95 percent confidence. Passing that many trials is not, by itself, operational assurance: representativeness, stationarity, hazard coverage, stopping rules, and configuration control still matter. Figure 2 visualizes the separate pure-series illustration and its assumptions.

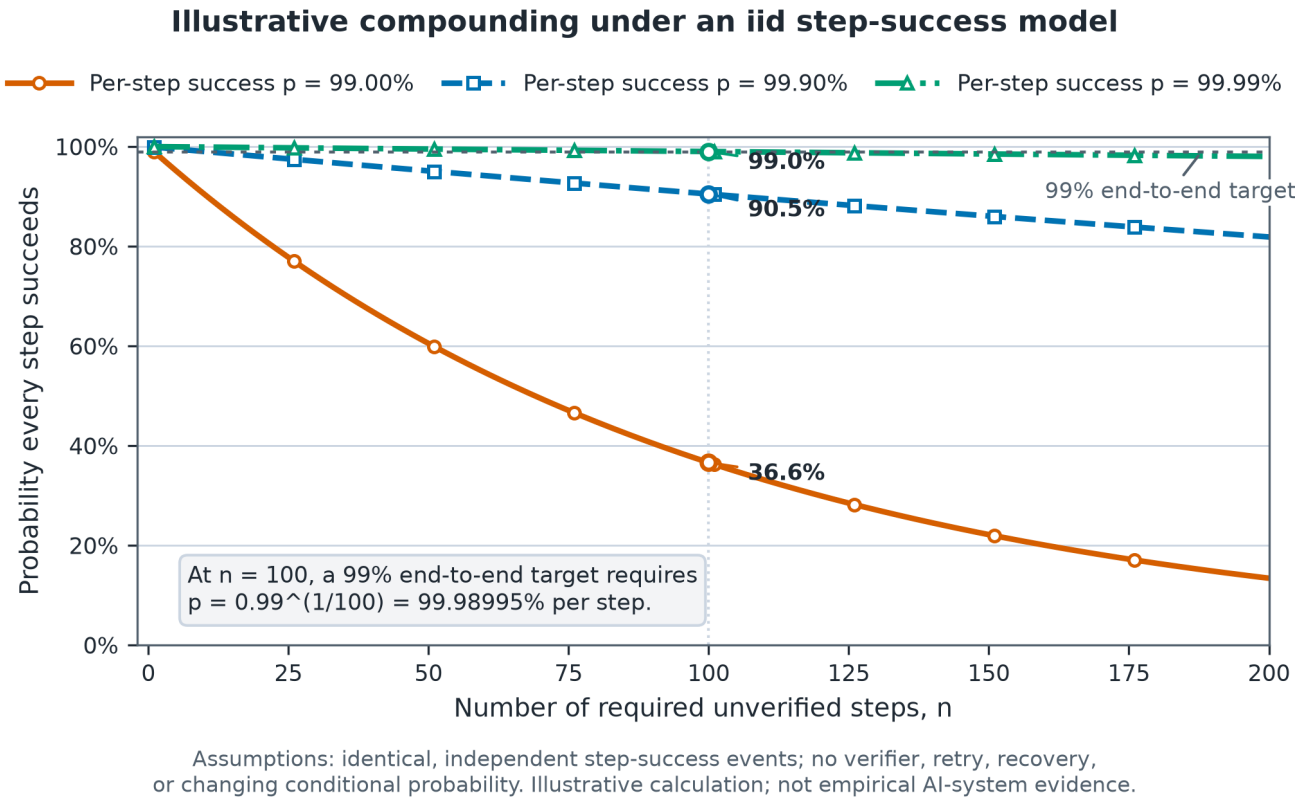


Figure 2. Illustrative compounding for a pure series workflow under identical, independent step-success events. Curves show p^n for p = 0.99, 0.999, and 0.9999 over 1–200 required unverified steps. At n = 100, the corresponding all-step success probabilities are 36.6, 90.5, and 99.0 percent; a 99 percent end-to-end target requires p = 0.99^(1/100) = 99.98995 percent. The calculation assumes no verifier, retry, recovery, optional branch, or changing conditional probability and is not empirical evidence about any AI system. Author calculation; source data are included in the supplement.

6.3 Mean performance is not a safety case

Most leaderboards report a mean, pass rate, or aggregate score. Those values answer an important but incomplete question. They do not usually report the severity of errors, concentration of failures, confidence calibration, or the number of retries and tokens consumed. A system with 90 percent average accuracy could make harmless mistakes

uniformly, or it could be nearly perfect on routine items and fail systematically on the rare cases that carry the greatest consequence.

Risk is a function of probability and impact, but both terms are conditional on context. The same hallucination rate has different implications in brainstorming and legal citation. A model's tendency to over-act is more consequential when connected to a production cloud account than when operating in a sandbox. Evaluation should therefore stratify results by hazard class, authority, reversibility, and exposure rather than reporting one global average.

6.4 Benchmark saturation, contamination, and Goodhart effects

Benchmarks become less informative when they are widely published, embedded in training corpora, repeatedly optimized, or reduced to a familiar format. Data contamination can inflate scores through direct or near-duplicate exposure, and even rephrased items may preserve enough structure to create exposure effects (Yang et al., 2024). Even without literal leakage, developers and researchers learn the benchmark's structure and tune prompts, tools, and post-training accordingly. Once a measure becomes a target, improvement may concentrate on the measure rather than the underlying construct.

The response has been a succession of harder, broader, or more dynamic evaluations: MMLU to MMLU-Pro and more difficult expert tests, static coding sets to repository tasks and live issue collection, visual question answering to multimodal expert reasoning, and isolated tool calls to interactive computer environments (Wang et al., 2024; Zhang et al., 2025). This succession is evidence of progress and of measurement decay. It prevents a simple chart from one benchmark in 2020 to a different benchmark in 2026 from being interpreted as a continuous intelligence scale.

Dynamic and live benchmarks reduce exposure but create new issues. They are harder to reproduce, may change over time, and can place evaluation power in the hands of a small organization. Private tests reduce contamination but limit public audit. The correct design is a portfolio: stable public tests for reproducibility, hidden or time-separated tests for generalization, adversarial tests for security, and field evidence for operational transfer.

6.5 Distribution shift and underspecification

Machine-learning systems can fit multiple functions that perform similarly on training and test data but behave differently after deployment. This problem, often described through underspecification and shortcut learning, is central to reliability. If a hospital model relies on institution-specific coding patterns, a vision model relies on background texture, or a language model relies on superficial wording, the benchmark may not reveal the dependency until the environment changes (Geirhos et al., 2020; Koh et al., 2021).

Foundation models reduce some forms of narrow overfitting by training on broad data, but breadth does not eliminate shift. New jurisdictions, adversarial users, uncommon combinations of known concepts, changing software interfaces, and rare physical events all create new distributions. The model may also have broad knowledge but lack calibrated evidence about whether the present case falls within that knowledge.

A dependable system needs explicit domain boundaries, shift detection, abstention criteria, and post-deployment monitoring. "Works on our sample" is not equivalent to "has an established operational envelope."

6.6 Calibration, uncertainty, and abstention

A reliable system must communicate when its output should not be trusted. Neural networks can be miscalibrated even when accuracy is high (Guo et al., 2017). Language models sometimes contain information useful for predicting whether their own answers are correct, but verbal confidence and sampled probabilities are sensitive to prompting and do not guarantee reliable calibration (Kadavath et al., 2022). Calibration should be scored with a rule appropriate to the forecast and decision, and selective systems should report risk as coverage changes rather than presenting abstention as cost-free (Geifman & El-Yaniv, 2019; Gneiting & Raftery, 2007).

Farquhar et al. (2024) found semantic entropy useful for detecting confabulations on the evaluated question-answering tasks. Consistency signals, retrieval scores, verifier outcomes, and ensemble disagreement can add evidence, but no indicator reviewed here established sufficient calibrated coverage across all considered domains and decision losses.

Calibration must be evaluated against the exact decision and loss function, and abstention must lead to a tested fallback rather than an ambiguous warning.

The human interface is part of this control. Systems that present fluent answers, hidden uncertainty, and a default action can induce automation bias. Users may accept incorrect recommendations or fail to monitor automation after prolonged success (Bainbridge, 1983; Parasuraman & Riley, 1997). Human-in-the-loop is not a control unless the human has enough information, time, authority, and skill to detect the relevant failure.

6.7 Long context is not persistent, correct state

A model's context window describes the maximum input capacity, not the quality of state management. Long tasks require the system to preserve goals, constraints, decisions, evidence, and environmental changes across many actions. Compression and summarization can discard details. Retrieval can return an obsolete record. Generated memory can convert an earlier inference into a later "fact."

Reliable state should therefore be externalized in typed, authoritative structures where possible. Plans should identify preconditions and postconditions. Tool results should be logged. Critical facts should point to sources. Checkpoints should support rollback. The model may propose updates, but it should not silently rewrite authoritative state.

6.8 Verification gaps in open-ended tasks

Verification is easiest when a correct result can be computed or tested. A compiler, theorem prover, unit test, schema validator, or physical measurement can reject many wrong outputs. For the open-ended examples considered here—strategy, intelligence analysis, policy advice, and scientific interpretation—this review did not identify a generally applicable, complete automatic verifier. Reviewers must assess evidence, assumptions, and alternative explanations.

This creates an asymmetry. Generators can produce plausible candidates at high speed, while verification remains expensive and domain-specific. Increasing generation volume without increasing verification capacity can worsen the signal-to-noise ratio. The appropriate productivity metric is accepted, correct, decision-relevant output per unit of total human and machine effort, not tokens generated or drafts completed.

7. Capability–Assurance Maturity Framework

7.1 Maturity levels

The proposed framework assigns a capability level (C0–C5) and an assurance level (A0–A5) to a specific function in a defined environment. It is an unvalidated author-developed decision aid, not a consensus standard, certification scheme, or psychometric instrument. Application requires documented scoring anchors, row-level evidence, independent review, and explicit uncertainty.

Table 4. Capability and assurance maturity levels

Capability level	Capability evidence	Assurance level	Assurance evidence and controls
C0	Claimed or conceptual; no adequate demonstration	A0	No defined requirements, hazards, or evidence
C1	Demonstrated on selected examples or a narrow benchmark	A1	Preliminary testing; assumptions and failure modes incompletely characterized
C2	Repeated under a controlled test distribution; independent reproduction recorded separately	A2	Documented evaluation, version identification, basic security controls, and known limitations
C3	Useful in a bounded operational workflow with supervision	A3	Operational monitoring, accountable roles, access control, fallback, incident handling, and change control
C4	Dependable within a defined operating context under measured material variation	A4	Independent validation, hazard-conditioned reliability, adversarial testing, fail-safe behavior, auditability, and evidence-backed risk treatment
C5	Adaptive and highly reliable across the specified material variation	A5	Continuous assurance, strong containment, rare-event evidence where applicable, robust governance, and validated operation under change

C5 is not synonymous with human-level general intelligence, and an A-level is always consequence- and use-specific. A narrow function could receive a high C/A pair while a cognitively broader system receives a lower assurance level. A rating does not imply zero risk; it records the evidence and controls available for a specified decision.

7.2 Assessment dimensions

A maturity assessment should cover at least ten dimensions:

Requirement validity: the intended function and prohibited outcomes are testable.

Evaluation validity: test tasks represent the operational distribution and hazards.

Reliability distribution: results include variance, tail behavior, and failure severity.

Generalization: performance under time, location, user, data, and environmental shift is measured.

Calibration and abstention: uncertainty leads to appropriate fallback.

Security: data, prompts, tools, models, and supply chains are protected under a threat model.

Authority: identity, permissions, approvals, and separation of duties are enforced outside the model.

Observability: actions, evidence, state changes, and configurations are traceable.

Recovery: failures can be detected, contained, reversed, and learned from.

Lifecycle control: model, data, prompt, tool, and policy changes trigger re-evaluation.

An organization should not average these dimensions into a passing score if one is a critical gate. For example, high task performance cannot compensate for unrestricted administrative credentials, and excellent logging cannot compensate for a system that has no safe state after a physical control error.

7.3 Domain assessment at the 2026 cutoff

Table 5 records an author synthesis, not a measured population statistic or universal rating. Each row assumes the stated context; an implementation may warrant a materially different assignment. The ranges express evidence uncertainty and configuration variation, and the supplementary claim audit records the principal evidence boundary for each row.

Table 5. Indicative capability and assurance maturity by capability–context pair (author assessment)

Capability–context pair	Capability level	Assurance level	Evidence boundary
Routine text drafting and transformation with accountable review	C3–C4	A2–A3	Scaled use and reversible review workflows; accuracy remains task- and reviewer-dependent
Retrieval-assisted enterprise question answering over governed sources	C3	A2	Utility depends on retrieval quality, citation checking, access control, and corpus governance
High-stakes open-ended advice without independent verification	C3	A1	Fluent generation does not establish factual, causal, or decision reliability
Speech transcription in specified languages and acoustic conditions	C4	A2–A3	Strong bounded performance; shift, speaker, privacy, and record-integrity limits remain
Code completion and bounded implementation assistance	C3–C4	A2–A3	Utility is supported in selected tasks when tests and review are available; security and maintainability vary
Autonomous repository maintenance across unfamiliar codebases	C2–C3	A1–A2	Benchmark progress is material, but task validity, tail behavior, and operational acceptance remain limited
Short digital workflows in controlled applications	C3	A2	Repeatable successes are configuration- and interface-dependent
Open-ended, multi-day digital agency with privileged tools	C2	A1	Public evidence does not establish high-reliability, contained operation

Capability–context pair	Capability level	Assurance level	Evidence boundary
Document and image analysis with human confirmation	C3–C4	A2	Bounded utility with grounding, privacy, and distribution-shift risks
Scientific candidate generation with an exact formal or experimental verifier	C3–C4	A2–A3	Evidence is strong in selected domains; the verifier and experimental pipeline define the boundary
Open-ended autonomous scientific research	C1–C2	A1	Demonstrations do not establish dependable end-to-end research autonomy
Foundation-model addition to an otherwise structured industrial cell	C1–C2	A1	Conventional robot-cell assurance does not transfer automatically to the learned component
General-purpose robotics in unstructured environments	C2	A1	Demonstrated transfer remains bounded; physical reliability evidence is limited
Named automated-driving system inside a specified operational design domain	C4	A3	Substantial field evidence can support a bounded claim; system, geography, weather, and governance matter
Autonomous cyber actions on production systems	C2	A1	Capability is material; adversarial uncertainty and side effects limit assurance

Within these author-assigned rows, capability usually sits above assurance, especially as authority and environmental openness increase. This is a cross-sectional ordinal pattern in the reviewed cases—not a measured rate of change, proof of a universal gap, or evidence that the difference is widening over time. Narrow functions with strong external verification or extensive domain engineering are the principal counterexamples.

8. Root Causes of the Gap

8.1 Objective ambiguity and proxy optimization

AI systems optimize training objectives that are proxies for intended behavior. Next-token prediction, preference scores, benchmark rewards, and task-completion metrics capture only part of what users value. When the proxy is incomplete, a capable optimizer can satisfy the measured objective while violating unstated constraints. Specification gaming in reinforcement learning is a long-standing example, and agentic systems enlarge the space of possible workarounds because they can manipulate tools, files, tests, and environmental state (Krakovna et al., 2020).

Preference-based post-training improves behavior on sampled interactions but inherits limitations from the labelers, prompts, and reward model. Casper et al. (2023) catalogued open problems in reinforcement learning from human feedback, including scalability, bias, reward misspecification, and the difficulty of supervising behavior beyond human evaluation. These are not arguments that post-training fails. They show that behavioral alignment on a test set is not a complete specification.

8.2 Data provenance and representation

Foundation models learn from heterogeneous data whose origin, license, quality, recency, and duplication are often incompletely documented. Datasheets for datasets and model cards were proposed to make the assumptions and intended uses of data and models explicit (Gebru et al., 2021; Mitchell et al., 2019). At frontier scale, documentation remains difficult because data are assembled from many sources and transformed through filtering, deduplication, synthetic generation, and post-training.

Poor provenance affects factuality, bias, intellectual-property risk, and evaluation contamination. It also obstructs root-cause analysis. When a model produces a harmful association or memorized passage, investigators may be unable to determine which data created the behavior. Synthetic data can extend training and target difficult skills, but recursive use of generated data can reinforce errors, reduce diversity, or make provenance even less transparent if quality controls are weak.

8.3 Distribution and environment mismatch

A benchmark is a sample from an evaluation distribution. An operation is a process that changes over time. Users adapt, adversaries probe, software interfaces change, sensors degrade, and institutional policies evolve. The system's actions can also change the environment it later observes. This feedback makes deployment fundamentally different from independent test items.

The mismatch is especially severe for agents. A wrong answer in a benchmark ends the item. A wrong action in an environment may delete information, alter permissions, trigger an alert, or make later observations misleading. The resulting failure is path dependent. Evaluation must therefore include stateful trajectories, side effects, and recovery, not only final task completion.

8.4 Hidden state, memory, and model opacity

Modern models encode behavior in high-dimensional representations that remain only partly characterized. Published mechanistic-interpretability studies have identified useful circuits and feature representations, including induction heads and sparse features, but the literature reviewed here did not establish a method that provides a complete, stable, causal explanation of a frontier system's behavior across all relevant inputs (Elhage et al., 2021; Olsson et al., 2022; Bricken et al., 2023; Templeton et al., 2024).

This limitation matters for assurance because testing cannot exhaust an enormous input space. Interpretability could, in principle, identify dangerous latent objectives or verify that a system uses the intended evidence. Current tools are valuable for research and targeted audits but are not a standalone safety proof. Features may be nonunique, incomplete, or difficult to connect to system-level behavior. Behavioral testing, causal intervention, architectural constraints, and monitoring remain necessary.

8.5 Coupling capability to authority

A model that can recommend an action presents a different risk from a model that can execute it. Agent frameworks often couple language-model output to credentials, APIs, browsers, code interpreters, and file systems. This changes the security boundary. Prompt injection becomes a potential command injection path; retrieved text can alter behavior; compromised tools can return malicious observations; and the model may act as a confused deputy with more authority than the requester.

The security problem is structural. In many language-model applications, untrusted natural-language content and authorized instructions enter the same inference context, and the model may not reliably preserve the intended separation. System prompts and refusal training influence behavior but do not provide the isolation properties of an independently enforced security boundary. Deterministic policy enforcement, least privilege, typed interfaces, and separation of duties should therefore exist outside the model.

8.6 Evaluation and deployment incentives

Organizations benefit from demonstrating capability quickly. Reliability evidence takes longer, exposes failures, and can delay deployment. Benchmarks reward visible scores; incident reporting can create legal or reputational cost; independent evaluation may require access that developers are reluctant to provide. These incentives produce an evidence imbalance even when all actors are acting rationally within their local objectives.

The compute divide compounds the problem. Frontier developers control the models, infrastructure, and often the first evaluations, while independent researchers may lack sufficient access to reproduce findings or test rare risks (Besiroglu et al., 2024). Public APIs enable some evaluation but can change silently, restrict logging, or hide system details. Assurance therefore requires institutional mechanisms for protected third-party access, regulator testing, reproducible snapshots, and disclosure of material changes.

9. Security, Safety, and Assurance Architecture

9.1 Threat model for AI-enabled systems

Security analysis must distinguish accidental model error from deliberate influence. The same architecture can fail through either path. An AI-enabled system processes training data, prompts, retrieved content, model outputs, tool responses, memory, and telemetry. Each can be manipulated or can carry an error into a later stage. The NIST adversarial-machine-learning taxonomy provides a useful common vocabulary for attack stages and mitigations, but it is not evidence that any listed mitigation is complete or effective in every deployment (Vassilev et al., 2025).

An illustrative baseline threat model should include at least:

Instruction-channel manipulation: direct and indirect prompt injection, malicious retrieved content, tool output injection, and cross-context instruction confusion.

Data integrity attacks: training-data poisoning, retrieval-corpus poisoning, manipulated feedback, synthetic-data contamination, and corrupted memory.

Credential and authority abuse: overprivileged agents, token theft, confused-deputy behavior, unauthorized delegation, and inability to attribute an action to a human, model, or service identity.

Model and supply-chain compromise: malicious weights, unsafe fine-tunes, dependency compromise, model substitution, insecure serialization, and untracked provider updates.

Confidentiality attacks: prompt leakage, data exfiltration through tools or outputs, memorization, membership inference, model inversion, and exposure through observability systems.

Availability and economic attacks: denial of service, context flooding, recursive tool use, token exhaustion, and intentionally expensive planning paths.

Evaluation manipulation: benchmark contamination, grader exploitation, test awareness, and reward hacking.

Monitoring evasion: deletion or corruption of logs, action outside approved interfaces, steganographic communication, and behavior that changes when evaluation is detected.

Universal adversarial suffixes demonstrated that aligned language models could be induced to produce disallowed outputs through optimized inputs, including transfer across models (Zou et al., 2023). Indirect prompt-injection research showed that malicious instructions embedded in content could compromise applications that connected language models to external data and tools (Greshake et al., 2023). A 2026 public competition supplied large-scale adversarial benchmark evidence for indirect prompt injection under the competition protocol; it did not estimate attack prevalence in fielded systems (Dziemian et al., 2026; National Institute of Standards and Technology, 2026c). These findings establish that conversational alignment is not a security boundary.

9.2 Goal misgeneralization, deceptive policies, and the limits of behavioral tests

Goal misgeneralization describes a system that remains competent after distribution shift while pursuing a learned objective different from the intended one (Langosco et al., 2022). The problem differs from ordinary capability failure: an agent can retain task competence while optimizing the wrong criterion. Sleeper-agent experiments provided constructed examples in which conditional policies designed to be deceptive persisted through several safety-training procedures (Hubinger et al., 2024). Those experiments do not establish that deployed frontier systems contain autonomous deceptive goals. They show, within the studied setups, that failure to observe misbehavior on a training distribution does not logically exclude conditional behavior elsewhere.

Behavioral evaluation remains essential, but its evidence must be interpreted statistically. Failure to observe a rare behavior in a finite test does not establish absence. Preprint evidence indicates that some language models can distinguish evaluation-like prompts under studied conditions, so evaluation-cue awareness is a plausible confound rather than a demonstrated universal behavior (Needham et al., 2025). Stronger evaluations randomize tasks, conceal some acceptance criteria, create opportunities for prohibited behavior, and test whether monitoring itself can be evaded. Even then, assurance should not rest on a claim that a model’s internal objective has been proved benign.

9.3 Externalize policy and authority

A defensible reference architecture treats the model as an untrusted but capable proposal component. Policy, authorization, and irreversible control remain outside it. Figure 3 illustrates this author-proposed pattern; it is not a validated reference implementation.

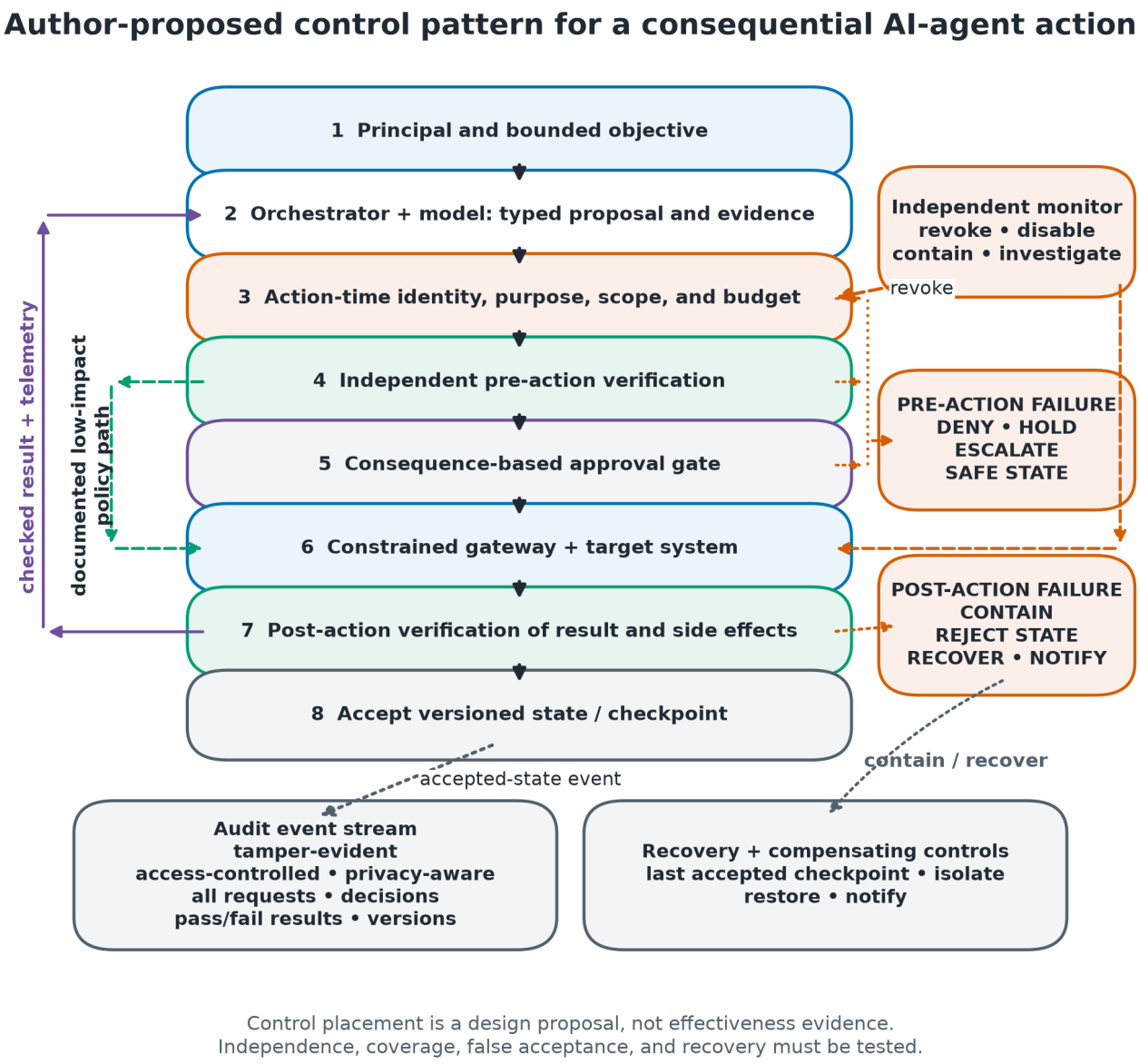


Figure 3. Author-proposed assurance architecture for an AI agent. A requester's bounded objective enters an orchestrator that asks the model for a typed proposal and evidence. Identity and policy enforcement authorize the specific principal, agent, action, resource, purpose, scope, and budget. An independently designed pre-action verifier evaluates preconditions, schemas, policy, and available semantic evidence before a consequence-based approval gate permits a constrained gateway to act on the target system. Required-control failure denies, holds, or escalates the action. Post-action verification checks results and side effects before state is accepted; checked telemetry can inform a subsequent proposal. Separate monitoring, tamper-evident audit, versioned state, and recovery controls support revocation, reconstruction, and response. The diagram specifies proposed control placement, not proven effectiveness.

The architecture contains several separations.

First, the orchestrator converts a model proposal into a typed action request; free-form text does not execute directly. Second, identity and policy enforcement evaluate the requester, agent instance, tool, resource, action, purpose, and context, issuing narrowly scoped, time-bounded authorization rather than standing model credentials. Third, a verifier checks preconditions, schemas, business rules, security policy, and, where possible, semantic evidence. Fourth, high-

impact or irreversible actions require an approval gate or a separately governed control. Fifth, tools run through a constrained gateway with quotas, network policy, and transaction boundaries. Results and telemetry return to the orchestrator. A separate, versioned state store and tamper-evident, access-controlled, privacy-aware audit trail record inputs, evidence, model and policy versions, proposals, approvals, tool results, and recovery actions.

This architecture does not eliminate model error. If implemented and tested effectively, it can convert some errors into rejected requests, contained experiments, or reversible transactions and can create evidence for incident analysis. The design applies established security and systems-engineering principles: least privilege, complete mediation, separation of duties, fail-safe defaults, zero-trust access decisions, defense in depth, and independent protection layers (Ross et al., 2022; Rose et al., 2020; Saltzer & Schroeder, 1975).

9.4 Agent identity and authorization

Enterprise identity systems were designed primarily for humans, services, and workloads with relatively stable functions. AI agents can be created dynamically, delegate tasks, select tools, and act on behalf of multiple principals. Treating every agent as a conventional user account or giving a shared orchestrator broad privileges creates governance and blast-radius problems.

An agent authorization model should bind every action to:

the human or system principal that initiated the objective;

the specific agent and model configuration;

the delegated purpose and time limit;

the tool and resource being accessed;

the maximum data sensitivity;

the permitted action type;

the required approval level;

the budget for compute, tokens, transactions, or physical actuation; and

the evidence and logs required for completion.

Capability-based or attribute-based authorization can implement these constraints. Delegation should be attenuating, meaning that a child agent receives no more authority than the parent and usually less. The system should prevent an agent from granting itself new permissions, altering its monitor, approving its own output, or accessing acceptance tests that it is optimizing against. NIST's February 2026 Agent Standards Initiative, the NCCoE draft concept paper, and NIST's May 2026 summary of responses to its agent-security request for information identify interoperability, identity, delegation, and secure operation as active technical and standardization problems. They are an initiative, a draft project concept, and a stakeholder-response summary—not completed agent-security standards or independent validation of the reported threats (National Cybersecurity Center of Excellence, 2026; National Institute of Standards and Technology, 2026a; Riggs et al., 2026).

9.5 Verification, monitoring, and rollback

Verification should be matched to output type. Code can be compiled and tested. Database actions can be checked against schemas, row limits, and transaction policies. Infrastructure changes can be planned, simulated, reviewed, and applied through policy-as-code. Analytical claims can require citations to approved evidence. Physical actions can be constrained by safety envelopes and independent interlocks.

Monitors should be independent enough that one model error does not corrupt both action and oversight. Independence should be evaluated across implementation, training data and model lineage, prompts and context, evidence sources, credentials, operators, and failure modes. Diversity can include deterministic rules, separately engineered models, anomaly detectors, and human review. A second language model is not automatically independent if it shares training

data, architecture, prompts, evidence, or failure modes with the first. Monitor effectiveness should be measured through seeded faults and adversarial attempts to bypass detection.

Recovery must be designed before autonomy is granted. Where operations are reversible, digital agents should use version control, snapshots, staged deployment, canary environments, and transactional APIs. Physical systems need independently enforced safe states, emergency stops, speed and force limits, protected zones, and recovery procedures appropriate to the hazard analysis. Some actions, including disclosure of secrets, completed transfer of funds, or physical harm, are not fully reversible. Those actions require stronger preventive controls, separate authorization, and compensating response measures rather than a promise of rollback.

9.6 Safety cases for AI

A safety or assurance case is a structured argument that claims are supported by evidence; ISO/IEC/IEEE 15026-2 defines assurance-case concepts and structure but does not supply an AI application's claims, evidence, or acceptance criteria (International Organization for Standardization, International Electrotechnical Commission, & Institute of Electrical and Electronics Engineers, 2022). For AI, the top-level claim should be specific: the system is acceptably safe and secure for a defined function, operational design domain, version, and authority. Evidence can include benchmark results, field tests, red-team findings, reliability analysis, formal verification of wrappers, access-control tests, human-factors studies, incident history, and monitoring performance.

The case should explicitly record defeaters: evidence or conditions that would invalidate the claim. A material model update, new tool, expanded user population, changed data source, higher privilege, or new operational environment may require reassessment. This is essential for continuously updated services. Certification of a model name without configuration and change control is insufficient.

NIST's Artificial Intelligence Risk Management Framework organizes AI risk activities around governing, mapping, measuring, and managing risk, and the Generative AI Profile adds risks specific to generative systems (Autio et al., 2024; Tabassi, 2023). ISO/IEC 23894 provides AI risk-management guidance, while ISO/IEC 42001 defines an AI management-system standard (International Organization for Standardization, 2023a, 2023b). The European Union's Artificial Intelligence Act adds legally binding, risk-based obligations and provisions for general-purpose AI (Regulation (EU) 2024/1689, 2024). As of August 28, 2026, general-purpose AI obligations had applied since August 2, 2025, and Commission enforcement powers and specified transparency rules applied from August 2, 2026. Regulation (EU) 2026/1744, in force from July 27, 2026, changed the principal high-risk-system application dates to December 2, 2027 for Annex III systems and August 2, 2028 for Annex I product systems (Regulation (EU) 2026/1744, 2026; European Union, 2026). These frameworks create useful organizational structure. They do not supply application-specific reliability thresholds or prove that a system is safe. Evidence must still be generated for the intended use and version.

9.7 Consequence-based control profiles

The rigor of assurance should increase with consequence and autonomy.

Table 6. Illustrative consequence-based control profile (author proposal)

Class	Typical use	Permitted model role	Minimum controls
C0: negligible	Ideation, low-impact formatting, private experimentation	Generate freely within policy	Basic privacy, user notice, and version tracking
C1: limited	Routine drafting, internal search, reversible productivity tasks	Recommend or execute reversible actions	Source access control, logging, user review, rollback, quality sampling
C2: material	Customer communication, software changes, business operations	Execute bounded actions under policy	Typed tools, least privilege, test and approval gates, monitoring, incident process
C3: serious	Security response, financial movement, legal or clinical support, infrastructure administration	Recommend; execute only predefined low-risk actions	Independent verification, dual control, strong identity, sandboxing, adversarial testing, recovery exercises

Class	Typical use	Permitted model role	Minimum controls
C4: catastrophic potential	Weapons, life-critical control, large-scale infrastructure, irreversible public safety decisions	No unconstrained model authority	Deterministic safety envelope, independent interlocks, formal evidence where feasible, accountable authorization, continuous assurance

The table is neither a legal classification nor a consensus standard. It is an author-proposed engineering profile: authority should not be granted merely because a model demonstrates high average capability. The greater the consequence and irreversibility, the more the system should rely on controls that do not share the model's failure modes.

10. Data, Compute, Energy, and Infrastructure Economics

10.1 Scaling is an economic and systems question

Scaling laws made compute allocation a central strategic variable, but the relevant objective is no longer training loss alone. Organizations pay for data acquisition and cleaning, training, post-training, evaluation, inference, retrieval, observability, human review, security, and incident response. Agentic systems can multiply inference cost through planning, retries, tool calls, and verifier models. A smaller model with retrieval and a strong verifier may produce lower total cost and higher reliability than a frontier model used without structure.

Training and inference costs also have different optimization regimes. A model trained once and served at enormous volume may justify greater training compute to reduce per-query cost. A specialized model with limited demand may not. Under their fitted scaling-law assumptions and tested model range, Sardana et al. (2024) showed that expected inference demand changes the parameter and token allocation that minimizes training-plus-inference compute or cost. Test-time compute adds another variable: additional sampled reasoning can improve performance on some verifiable tasks, but marginal gains must be evaluated against latency, energy, and failure opportunities.

Procurement should therefore require total-system metrics, including accepted task cost, energy per verified result, human review time, latency at the required reliability, and cost of failed or rolled-back actions. Price per token is not a mission outcome.

10.2 Data constraints and synthetic data

High-quality human-generated data are finite, unevenly distributed, legally constrained, and increasingly entangled with model-generated content. Data-constrained scaling research indicates that repeated training data can continue to provide value for a time but with diminishing returns (Muennighoff et al., 2023). Synthetic data can target rare tasks, provide process traces, and generate adversarial examples. In recursively trained generative-model experiments, inadequate preservation of original-distribution data caused loss of tail information and reduced distributional diversity; that conditional result does not imply that all synthetic-data use causes collapse (Shumailov et al., 2024).

The critical resource is not raw token count. It is high-information data with provenance, rights, coverage, and an evaluator. For long-horizon agents, trajectories containing recovery, ambiguity, and failure may be more valuable than large volumes of successful short interactions. For safety, near-miss and incident data are essential but difficult to share. Research infrastructure should support protected pooling, standardized taxonomies, and privacy-preserving analysis.

10.3 Compute concentration and independent evaluation

Frontier training requires capital, specialized accelerators, networking, power, cooling, and engineering expertise. In a 2024 preprint, Besiroglu et al. reported that compute disparity coincided with reduced representation of academic-only teams in compute-intensive topics, especially foundation-model research. The study supports concern about access disparity but does not by itself establish a mature causal consensus. Concentration can accelerate capability through focused investment, but it can also concentrate evidence production. The organizations with the greatest ability to develop models often control model access, training details, and the initial evaluations.

Independent scrutiny requires more than open papers. Evaluators need stable model versions, sufficient rate limits, appropriate logging, the ability to test dangerous behaviors safely, and protection for sensitive findings. National and

shared research-compute programs can support independent science, but access to closed frontier systems may still require confidential evaluation agreements or regulatory authority.

Open-weight models change the trade space. They improve inspectability, reproducibility, local control, and research access. They can also be fine-tuned to remove safeguards and deployed without provider monitoring. Closed systems can enforce access and updates but create dependency, reduce auditability, and change behavior without users controlling the version. “Open” and “closed” are not safety rankings. The appropriate choice depends on threat model, governance capacity, and mission.

10.4 Energy and physical infrastructure

Lawrence Berkeley National Laboratory’s 2024 bottom-up model estimated U.S. data-center electricity use at 176 terawatt-hours, or 4.4 percent of U.S. electricity, in 2023 and produced a 2028 scenario range of 325–580 terawatt-hours, or 6.7–12.0 percent (Shehabi et al., 2024). Its June 2026 update estimated a 2030 reference case of 649 terawatt-hours, or 11.8 percent, with compounded-uncertainty bounds of 521–843 terawatt-hours, or 9.5–15.3 percent (Smith et al., 2026). These are modeled estimates and scenarios, not metered national AI consumption or committed future load. Accelerated servers used for AI are a major modeled driver alongside conventional computing and other data-center services; the national share attributable specifically to generative AI remains unclear (U.S. Government Accountability Office, 2025a).

Energy impact depends on location and timing, not only annual consumption. Large new data-center loads can create local generation, transmission, interconnection, and water constraints; effects depend on grid mix, climate, cooling design, load profile, and whether supply additions occur. Efficiency gains may reduce energy per task while total use rises through demand. Reporting should distinguish training and inference, operational and embodied emissions, power and energy, average and peak load, and gross consumption versus additional system impact.

Semiconductor supply chains add strategic concentration. Advanced-compute systems depend on specialized design, leading-edge logic and memory fabrication, advanced packaging, materials, and manufacturing equipment. A December 2025 U.S. Government Accountability Office review reported that, as of 2022, about three-quarters of chips were manufactured and packaged in Asia, with leading logic and memory production especially concentrated in Taiwan and South Korea; concentration differs across supply-chain segments (U.S. Government Accountability Office, 2025b). Hardware constraints can shape access to capability and the resilience of critical services. Organizations deploying AI as mission infrastructure should plan for provider outage, accelerator scarcity, model portability, and degraded operation.

10.5 Technical debt and lifecycle cost

Machine-learning systems accumulate technical debt through data dependencies, hidden feedback loops, configuration, monitoring gaps, and changes in the environment (Sculley et al., 2015). Foundation-model applications add prompts, retrieval corpora, embedding models, vector databases, tool schemas, guardrails, and provider APIs. A change to any component can alter behavior.

The lifecycle baseline must therefore include model identifier, weights or provider version, system prompt, post-training adapter, retrieval index, data snapshot, tool versions, policy rules, and evaluation suite. Regression testing should focus not only on average accuracy but on prohibited behavior, high-severity hazards, subgroup performance, security, and human workload. An application that cannot reproduce the configuration associated with an incident cannot support mature assurance.

11. Sector Implications

11.1 Knowledge work and enterprise operations

The most defensible enterprise opportunity is task decomposition rather than job-level autonomy. AI can draft, classify, retrieve, compare, summarize, and propose actions. Humans and deterministic systems can verify evidence, approve

exceptions, and accept responsibility. Productivity depends on whether the workflow reduces total effort, not whether generation is faster.

Organizations should begin with high-volume tasks that have clear acceptance criteria, governed data, and low-cost rollback. They should measure accepted output, correction time, error severity, user calibration, and process cycle time. Pilots that measure only user satisfaction or tokens saved are insufficient. Scaling should be contingent on evidence that the control burden does not erase the benefit.

11.2 Software engineering and cybersecurity

Software development will continue to be an early domain for advanced agents because repositories, tests, and version control create a rich verification environment. The highest-value systems will combine code generation with issue understanding, test creation, static analysis, dependency intelligence, and maintainability review. Unattended merging into production should remain gated by consequence, coverage, and rollback.

Cybersecurity is both an application and a threat domain. AI can accelerate alert triage, query generation, malware analysis, detection engineering, threat hunting, and remediation planning. It can also assist vulnerability discovery, phishing, exploit adaptation, and evasion. Autonomous defensive actions face an adversarial environment in which observations may be planted and attackers may intentionally manipulate the model. Security agents need especially strong provenance, tool isolation, identity, and approval. The model should never infer authorization from natural-language content found in telemetry or a ticket.

11.3 Critical infrastructure and defense

Critical infrastructure and defense combine high consequence, long system lifecycles, heterogeneous sensors, intermittent communications, and adversarial conditions. Foundation models can support maintenance, intelligence fusion, logistics, cyber defense, planning, and operator assistance. They should not be placed directly in a safety or weapons-control loop without an assurance architecture appropriate to the hazard.

A mission system should preserve deterministic command authority, bounded operating modes, independent sensing where feasible, and a degraded mode that does not require the model. AI recommendations should carry provenance and confidence. Exercises should include poisoned telemetry, communications loss, model outage, contradictory data, and compromised tools. Operational testing must examine the human team, not only model accuracy.

11.4 Health, law, and finance

Health, legal, and financial systems are information-rich and therefore attractive for AI assistance. They are also domains in which plausible errors can create serious harm. Strong uses include documentation, coding, retrieval, administrative support, and second-look analysis under professional accountability. Higher-risk uses require validated populations, calibrated uncertainty, source evidence, and clear responsibility.

The central governance error is to label a human as “in the loop” without measuring whether that person can detect the model’s failures. Reviewers may be overloaded, defer to a confident system, or lack access to the evidence. Effective oversight requires workload limits, interface design, training, and audit of disagreement and override patterns.

11.5 Science and engineering

AI can increase the rate at which candidate hypotheses, designs, programs, and materials are generated. The bottleneck may shift to simulation, experiment, expert interpretation, and replication. Research programs should invest in machine-readable experimental provenance, automated laboratories with safety interlocks, formal verification, and negative-result capture. Generating more candidates is valuable only when the evaluation pipeline can distinguish useful novelty from noise.

11.6 Robotics and manufacturing

In manufacturing, generalist perception and language interfaces are more defensibly introduced first in constrained cells and supervised logistics than in unbounded physical operation. Safety-rated controls, guarded zones, force and speed limits, and independently enforced programmable safety functions should remain the protection layer.

Foundation models may improve flexibility but should not inherit the assurance rating of conventional industrial automation or replace independent interlocks. Expansion to less structured environments should proceed through explicit operational design domains and monitored exposure.

12. Prioritized Research and Engineering Agenda

12.1 Priority model

The agenda groups gaps into author-proposed priority tiers using mission impact, dependency leverage, security consequence, urgency, evidence deficit, and five-year tractability. The criteria were applied qualitatively; no empirical basis was found for assigning defensible decimal weights. Table 7 is therefore a portfolio decision aid, not a validated ranking instrument. Items within a tier are intentionally unordered.

Table 7. Proposed research and engineering priority tiers

Tier	Research or engineering gap	Difficulty	Rationale
I	Runtime verification and constrained autonomy	High	Converts some model proposals into bounded, testable, and rejectable actions
I	High-reliability long-horizon execution	Very high	Determines whether systems can complete consequential workflows rather than isolated tasks
I	Dynamic, adversarial, and field-based evaluation	High	Tests operational transfer, side effects, and recovery beyond static benchmarks
I	Agent identity, authorization, and containment	High	Limits authority and blast radius as tool-using systems scale
I	OOD robustness, calibration, and abstention	Very high	Addresses systematic failure under environmental change
I	Incident learning and reliability-data infrastructure	High	Enables evidence about rare failures, near misses, and control performance
II	Data provenance and contamination-resistant evaluation	High	Protects the validity of training, benchmarking, and factual outputs
II	Interpretability tied to assurance decisions	Very high	Could reveal mechanisms and policies that behavioral testing misses
II	Causal and world-model grounding	Very high	Supports intervention, science, planning, and physical reasoning under explicit assumptions
II	Persistent memory and continual learning with control	Very high	Supports long-lived systems without silent drift or corrupted state
II	Independent evaluation access and reproducibility	High	Reduces dependence on developer self-evaluation
III	Efficiency, energy, and resilient AI infrastructure	High	Determines affordability, scale, and strategic resilience
III	Human–AI team design and supervisory capacity	High	Oversight can fail under excessive workload, speed, or authority
III	Version-aware governance, acquisition, and standards	High	Aligns evidence and controls with continuously changing systems

12.2 Runtime verification and constrained autonomy

Runtime verification is a high-leverage nearer-term priority. Research should develop verifiers that can check model proposals against formal constraints, tests, policies, and environmental invariants. The goal is not a universal truth machine. It is to expand the set of actions for which unacceptable outcomes can be detected before commitment.

Important directions include proof-carrying or evidence-carrying outputs, typed action languages, independently generated tests, transaction-level safety predicates, policy-as-code for agent actions, and verifiers resistant to collusion with the generator. Benchmarking should measure verifier false acceptance, false rejection, coverage, and susceptibility to adversarial examples. Systems should expose when a result is unverifiable and reduce authority accordingly.

12.3 High-reliability long-horizon execution

Agent research should report reliability curves rather than one success rate. METR reports fitted 50 and 80 percent task-completion horizons; 90, 99, and 99.9 percent horizons are proposed here as mission-dependent extensions, not

established benchmark conventions or currently well-estimated quantities. Any reported horizon must be conditioned on task class, scorer, system configuration, inference budget, and failure severity. METR's June 2026 predeployment report illustrates that horizon estimates can be sensitive to suite saturation and modeling choices; it is provider-access evaluation evidence, not operational validation (METR, 2026a). Long-task studies should include interruptions, ambiguous state, partial tool failure, stale data, and opportunities for unsafe shortcuts.

Architectural research should emphasize hierarchical planning, explicit state, checkpointing, subgoal contracts, uncertainty propagation, self-repair, and independent verification. Agents should learn when to stop and escalate, not only how to continue. Recovery metrics should include mean time to detection, containment, rollback, and human resolution.

12.4 Live evaluation and incident science

Static benchmarks should be complemented by continuously refreshed tasks, time-separated data, private adversarial sets, and field evaluation. Evaluation organizations need governance that prevents leakage while preserving enough transparency for scientific scrutiny. NIST AI 800-2 is an initial public draft on automated language-model benchmark practice and should be treated as draft guidance, not a final consensus method (National Institute of Standards and Technology, 2026d). Results should identify model and system versions, prompts, tools, inference budgets, retries, and cost.

AI incident reporting needs a taxonomy spanning model error, data failure, security compromise, misuse, human-automation interaction, and organizational response. Near misses are particularly valuable because they reveal weak controls before catastrophic loss. Legal safe harbors or protected sharing may be required to overcome reporting disincentives.

12.5 Agent identity, authorization, and containment

Research should define agent identity and delegation protocols that scale to large, ephemeral fleets. Required properties include provenance to an accountable principal, attenuating delegation, revocation, purpose limitation, resource budgets, nonrepudiation, and separation between planning and approval. Authorization decisions should be explainable without relying on model-generated rationales.

Containment research should measure blast radius under credential compromise, prompt injection, tool compromise, and colluding agents. Sandboxes should be evaluated against realistic escape and exfiltration attempts. Organizations also need operational patterns for key rotation, agent retirement, emergency disablement, and forensic reconstruction.

12.6 Robustness, uncertainty, and causal grounding

Robustness research should move beyond average OOD accuracy toward hazard-conditioned performance and calibrated abstention. Evaluations should hold out geography, time, institutions, languages, interfaces, and combinations of conditions. Models should be rewarded for identifying unsupported assumptions and requesting information when the cost of an incorrect action exceeds the cost of delay.

Causal research should distinguish retrieval of known causal statements from inference under explicit assumptions. Language models can help formulate causal graphs and interpret results, but intervention claims require identification strategies, data, and domain knowledge (Kıcıman et al., 2023; Jin et al., 2023). Coupling foundation models with structural causal models, simulators, and physical constraints is more defensible than treating fluent explanation as causal proof.

12.7 Interpretability that changes assurance decisions

Interpretability research should be evaluated by causal and operational criteria. Does the identified feature predict behavior across inputs and model versions? Does intervention on the mechanism change the behavior as predicted? Can the method detect a hidden policy before behavioral testing does? Does it reduce the cost or improve the coverage of an assurance decision?

Readable visualizations are not enough. A successful research program would connect internal mechanisms to test generation, monitor design, and control. Until then, interpretability should be one source of evidence among several.

12.8 Human supervisory capacity

The research community measures model task horizons more often than human oversight horizons. A supervisor may monitor one agent effectively but fail when responsible for dozens of concurrent agents that act quickly and present plausible explanations. Research should quantify supervisory span, alert fatigue, intervention latency, skill decay, and the effects of automation on situation awareness.

Interfaces should present evidence, uncertainty, planned actions, and consequence, not only a conversational rationale. Organizations should test whether operators can detect seeded errors and whether the system defers appropriately. Human review that cannot catch failures should not be counted as risk reduction.

12.9 Reproducibility and independent access

Frontier evaluations should preserve model snapshots or equivalent version identifiers, prompts, tool schemas, inference settings, and analysis code. Where weights cannot be released, qualified independent evaluators should receive stable access under terms that permit publication of methods and material findings. Regulators may require confidential access for dangerous-capability testing.

Public research infrastructure should support evaluation, not only training. Shared environments, cyber ranges, robot testbeds, and secure data enclaves can make operational evidence available to researchers who cannot build frontier models.

13. Technology Roadmap and Evidence-Bounded Scenarios

13.1 Two-year horizon: 2028

The 2028 base proposition is continued expansion of bounded agents. Systems may perform longer software, research, administrative, and analytical workflows, while higher-value deployments rely on external state, policy enforcement, tests, approvals, and recovery. Small groups of agents may divide planning, execution, critique, and verification, although the independence of those roles remains limited when they share models, data, evidence, or prompts.

Under this proposition, integration and governance may constrain adoption more than basic access to capable models. Organizations that treat agents as privileged workloads could obtain safer, more repeatable scaling than organizations that connect free-form interfaces directly to broad credentials. Regulatory and procurement requirements may place greater emphasis on evaluation evidence, incident processes, and model-change controls even where formal certification remains immature.

The upside proposition requires substantial gains in verifier effectiveness, calibration, and long-task reliability. If systems demonstrate high success on realistic multi-hour tasks with low side effects and manageable human intervention, adoption could accelerate into routine business operations. The downside proposition is a sequence of security and operational incidents involving overprivileged agents, poisoned context, or hidden workflow errors. Such incidents could slow grants of authority and shift spending toward controls without necessarily stopping capability progress.

This review assigns the 2028 proposition greater qualitative plausibility than the longer-horizon propositions because many component technologies already exist. The rate at which organizations can produce credible operational evidence remains uncertain; this is an analytical judgment, not a calibrated probability.

13.2 Five-year horizon: 2031

The 2031 base proposition is domain-specific autonomous operation supervised through policy and exception management. Software maintenance, laboratory workflows, logistics, customer operations, and security analysis may be performed by agent systems for longer intervals within defined boundaries. Model capability could become less differentiating than access to governed data, workflow integration, verifiers, and incident history.

Under this proposition, mature organizations maintain AI assurance cases that link each autonomous function to evaluation, permissions, monitors, and change triggers. Model updates are treated as safety- or security-relevant component changes. Dynamic benchmarks and protected incident exchanges could provide stronger operational evidence than current public leaderboards.

The upside scenario requires progress in high-percentile reliability, persistent state, OOD detection, and causal world models. If 99 percent task horizons expand by orders of magnitude rather than only median horizons, agents could manage complex projects with limited intervention. The downside scenario is an assurance plateau: models become more capable on tests, but correlated failures, adversarial manipulation, and supervisory overload prevent high-consequence autonomy. In that world, AI still produces large economic value as an assistant and generator, but claims of full workforce substitution remain overstated.

The 2031 proposition has lower qualitative confidence than the 2028 proposition. Five years permits substantial engineering progress but also gives deployment failures time to expose previously hidden constraints.

13.3 Ten-year horizon: 2036

The ten-year outlook has low confidence and should be treated as a decision space.

In the **assured-autonomy proposition**, research would succeed in combining capable learned models with robust state, runtime verification, causal and physical models, secure delegation, and continuous assurance. Systems could perform broad portfolios of digital and physical work within explicit legal, organizational, and technical boundaries. Human roles could shift toward objective setting, governance, exception handling, and accountability. The defining advance would not be error-free models, but system-level detection, containment, and recovery appropriate to the mission.

In the **capability-without-assurance proposition**, models would become substantially more capable but remain difficult to characterize under novelty, adversarial pressure, and long horizons. AI could transform work through coproduction and bounded automation while high-consequence authority remains restricted. Repeated incidents could follow where organizations confuse capability with assurance, and governance could become fragmented and reactive.

In the **constrained-scaling proposition**, data, energy, semiconductor, cost, or algorithmic limits would reduce the rate of frontier improvement. Innovation could shift toward specialized models, retrieval, formal methods, embodied data, and efficiency. Economic effects could remain material because diffusion and integration may continue even if frontier benchmark gains slow.

This review did not identify a scientifically defensible basis at the August 28, 2026 cutoff for assigning precise probabilities to human extinction, universal abundance, or a specific year for artificial general intelligence. Those propositions depend on unresolved definitions, technical breakthroughs, political choices, and feedback among capability, access, and control. The roadmap instead identifies observable conditions that could strengthen or weaken its propositions.

Table 8. Conditional roadmap propositions and evidence tests

Horizon	Base proposition	Conditions that strengthen it	Conditions that weaken it	Qualitative confidence
2028	Bounded agents expand in software and knowledge workflows under explicit controls	Improving multi-hour reliability, effective verifiers, constrained authority, and manageable intervention	Material incidents, weak verifier performance, or control burden that erases value	Moderate–high; analytical judgment, not probability
2031	Domain autonomy expands inside defined operational design domains	High-percentile reliability, controlled persistent state, calibrated abstention, and useful incident data	Assurance plateau, correlated failure, or supervisory overload	Moderate; analytical judgment, not probability
2036	Outcomes diverge among assured autonomy, bounded coproduction, and constrained scaling	Causal and physical grounding, secure delegation, continuous assurance, and resilient infrastructure	Persistent correlated failures, infrastructure constraints, or fragmented governance	Low

13.4 Leading indicators

The following indicators provide more decision value than general claims that “AI is getting smarter.”

High-reliability task horizon: proposed mission-dependent estimates of the duration and complexity at which systems achieve 90, 99, and 99.9 percent success on diverse realistic tasks, reported with uncertainty and sample-size limits. These are extensions beyond METR's reported 50 and 80 percent horizons.

Intervention burden: human minutes, approvals, and corrections per accepted task.

Failure severity distribution: not only error frequency, but the share causing irreversible, security, financial, or safety consequences.

OOD degradation: the change from in-distribution to time-, geography-, institution-, and adversary-shifted evaluation.

Calibration and abstention utility: whether confidence and refusal improve expected loss under actual decision costs.

Verifier effectiveness: false acceptance and false rejection for independent runtime controls.

Incident rate per autonomous action: normalized by exposure and consequence.

Rollback and recovery performance: time to detection, containment, restoration, and learning.

Reproduction lag: time between a developer claim and credible independent verification.

Benchmark freshness and contamination evidence: age, exposure, and leakage risk of major tests.

Energy and cost per verified outcome: total system cost, not price per token.

Supervisor span: number of agents or workflows one human can oversee while retaining failure-detection performance.

Authority concentration: the typical privilege, data access, and irreversible action available to deployed agents.

Assurance adoption: share of consequential deployments with change control, red teaming, incident reporting, and explicit operational design domains.

13.5 Falsification criteria

The base roadmap should be revised downward if, by 2028, independent evaluations show little improvement in high-reliability task horizons despite higher benchmark scores; if intervention burden remains constant or rises for complex work; if security incidents scale faster than autonomous actions; or if inference cost and energy prevent economically useful deployment outside a small number of firms.

The roadmap should be revised upward if multiple independent organizations demonstrate low-side-effect, high-percentile reliability on diverse multi-hour or multi-day tasks; if verifiers catch most consequential model errors without prohibitive false rejection; if agents maintain correct, auditable state across updates and interruptions; and if those results reproduce across model families and environments.

The long-term assured-autonomy scenario should be rejected if progress remains limited to mean benchmark performance while OOD degradation, latent-objective uncertainty, and supervisory overload do not improve. It becomes more plausible if causal intervention, mechanistic auditing, and runtime control produce evidence that generalizes across systems rather than succeeding only on curated demonstrations.

14. Discussion

14.1 The evidence does not support technological dismissal

The capability–reliability argument should not be misread as a claim that current AI progress is superficial. Broad pretrained systems have changed the cost of producing language, code, images, analysis, and candidate solutions. Scientific systems have generated results that can be checked independently. Speech and perception systems operate at scale. Digital agents can complete workflows that were beyond practical automation only a few years earlier. These are consequential advances.

The appropriate criticism is evidentiary, not ideological. The more capable a system becomes, the more valuable it is to measure correctly and control safely. Dismissing capability encourages unprepared deployment because institutions fail

to plan for rapid adoption and misuse. Exaggerating reliability creates a different failure by granting authority that the evidence does not support.

14.2 Scaling may continue without closing the gap

One counterargument is that the limitations described in this paper are temporary artifacts of current scale. Continued training, better post-training, and greater inference-time computation may improve robustness, reasoning, and planning. The evidence supports continued improvement as a plausible scenario. It does not support the stronger assumption that every assurance problem will disappear automatically.

Some gaps may scale favorably. Broader data can reduce narrow shortcuts, larger models can follow instructions more consistently, and test-time search can solve harder verifiable problems. Other gaps are created or amplified by capability. A more capable agent can search a larger action space for unintended solutions, exploit tools more effectively, and produce more persuasive errors. Long horizons create more opportunities for state divergence. Greater economic use increases exposure. External controls, evaluation science, and governance remain necessary even if model error rates fall sharply.

The relationship is therefore not “capability versus safety” as competing objectives. Reliable control is what allows capability to be used at greater authority and scale.

14.3 Engineering wrappers are necessary but not sufficient

A second counterargument is that conventional software engineering can wrap an unreliable model and solve the problem. This is partly correct. Typed tools, tests, least privilege, monitoring, and rollback can convert many model errors into safe failures. The strongest near-term recommendation of this paper is to use those mechanisms.

Wrappers are not sufficient when the task lacks a reliable verifier, when the model controls the evidence seen by the verifier, when the environment is physical and irreversible, or when the policy itself is ambiguous. A deterministic guardrail can enforce a transaction limit but cannot decide whether a strategic recommendation is causally sound. Assurance must combine architecture with domain evidence and accountable judgment.

14.4 Human oversight is a limited resource

A third counterargument is that humans can simply review AI outputs. Review is effective for low-volume tasks where errors are visible and reviewers have expertise. It becomes weaker as output volume, speed, and complexity increase. Automation can reduce operator engagement and situation awareness, while plausible explanations can mask incorrect premises.

Human oversight should be engineered like any other safety control. Its detection probability, workload, response time, and failure modes should be tested. The system should route cases according to consequence and uncertainty, not ask a person to approve every routine action until attention collapses. High-quality interfaces expose evidence and changes, not only prose.

14.5 Open and closed models create different assurance problems

Open-weight systems permit local testing, configuration control, and independent research. They can be integrated without sending sensitive data to a provider. Closed services may offer stronger centralized abuse monitoring, rapid security updates, and controlled access to dangerous capabilities. They also limit inspectability and can change without the operator controlling the model version.

Neither model is inherently dependable. Open deployment requires the operator to assume patching, access control, monitoring, and misuse prevention. Closed deployment requires contractual versioning, evaluation access, data protection, outage planning, and provider-risk management. Assurance depends on the complete governance and threat model.

14.6 Reliability standards need not freeze innovation

Standards are sometimes criticized for being too slow for AI. Static certification of a model snapshot would indeed become stale. The solution is not absence of standards; it is lifecycle-aware assurance. Standards can define evidence categories, material-change triggers, incident reporting, access control, and monitoring without prescribing one architecture.

A mature regime should permit rapid iteration inside a controlled envelope. Low-consequence uses can accept lighter evidence. High-consequence authority requires stronger gates. This proportional approach preserves experimentation while preventing the most consequential claims from resting on a demonstration.

14.7 The real unit of progress is verified autonomy

Benchmark scores will continue to be useful, but the decision-relevant frontier is verified autonomy: the amount of useful work a system can complete with bounded authority, acceptable intervention, and controlled failure. This unit combines capability, reliability, cost, and consequence.

Verified autonomy can grow through better models, better verifiers, narrower tasks, stronger tools, safer interfaces, or improved organizational design. It avoids the false choice between model-centric optimism and model-centric pessimism. The engineering objective is not to make a model appear intelligent. It is to deliver dependable mission outcomes.

15. Recommendations

15.1 For AI developers

Developers should report evaluation conditions in enough detail to support comparison: model version, date, prompt or system policy, tools, inference budget, retries, benchmark revision, and uncertainty. Results should distinguish public static tests from private or live tests. High scores should be accompanied by failure examples, calibration, and side-effect analysis.

Agent products should provide typed tool interfaces, scoped credentials, action previews, policy hooks, tamper-evident and access-controlled logs, budget controls, and recovery mechanisms appropriate to the action. Safety features should be testable by customers and independent evaluators rather than existing only as provider assertions. Model and policy changes should include material-change notes and regression evidence.

15.2 For enterprise and government adopters

Adopters should define the operational design domain before selecting a model. Requirements should specify permitted tasks, prohibited actions, data classes, authority, latency, reliability, fallback, and accountable roles. Evaluations should use the organization’s data and realistic adversarial conditions. Pilots should measure total process performance, including review and recovery.

The default deployment pattern for consequential agents should be zero trust: no standing broad credentials, no direct execution of free-form output, complete mediation, least privilege, approval for irreversible actions, and continuous monitoring. Acquisition language should require stable version identifiers, incident notification, evaluation access, data-use restrictions, portability, and exit plans.

15.3 For standards bodies and regulators

Standards should define a common vocabulary for capability evidence, operational transfer, incidents, material changes, and agent authority. Requirements should scale with consequence and autonomy. Regulators should avoid relying on a single model threshold or benchmark because system capability changes through tools, scaffolds, and deployment configuration.

Protected reporting and independent evaluation access are priorities. Without incident and exposure data, regulators will overreact to anecdotes or underreact to hidden patterns. Confidential testing can protect proprietary information while enabling credible oversight.

15.4 For research funders

Funding should prioritize shared evaluation infrastructure, secure agent testbeds, independent compute access, incident data, verifier research, and human supervisory studies. These are public goods that commercial incentives may underprovide. Fundamental work on causal models, interpretability, robust learning, and continual memory remains essential, but it should be connected to measurable assurance outcomes.

15.5 For researchers and publishers

Researchers should avoid treating provider labels as stable scientific entities. Papers should record exact model and API dates, prompts, tools, and sampling. Claims based on preprints or developer reports should be labeled. Negative results and failed replications should be publishable because they improve the reliability map.

Review articles should separate evidence of capability from evidence of deployment. A leaderboard table without benchmark validity, contamination risk, and operational context can create more confidence than knowledge.

16. Conclusion

During 2020–2026, broad models became materially more useful across language, code, perception, science, and action, and selected systems produced independently checkable or operationally measured results. The reviewed evidence does not support dismissing the period's progress as merely cosmetic, but it also does not establish one universal capability threshold.

The same evidence does not support unrestricted confidence. Capability remains uneven across tasks and conditions. Benchmark success can fail to transfer under distribution shift, long horizons, adversarial interaction, ambiguous objectives, or requirements for rare-event reliability. As models acquire tools and authority, errors become state changes rather than incorrect text. Assurance depends increasingly on identity, authorization, verification, monitoring, rollback, incident learning, and accountable human institutions.

The central synthesis is a capability–assurance asymmetry in many reviewed capability–context pairs: public evidence for useful task performance is often stronger than public evidence for dependable operation under high consequence, environmental variation, long horizons, or adversarial pressure. The review does not establish that this difference is universal or widening over time. It identifies the difference as an engineering and research agenda for using advanced AI with evidence commensurate to authority and consequence.

The most promising path does not wait for a perfectly aligned or fully interpretable model. It combines stronger models with externalized state, independent verification, bounded authority, operational design domains, calibrated abstention, and continuous evidence. Research should measure reliability curves and verified autonomy, not only average benchmark scores. Organizations should treat frontier models as high-capability but partially characterized components whose operational envelopes must be established empirically.

Whether the next decade produces broad assured autonomy, powerful bounded coproduction, or repeated cycles of capability and incident will depend less on one benchmark or one model release than on whether evaluation and assurance become first-class technologies. That is the capability–reliability gap, and closing it is the work that determines how much of AI's demonstrated potential can become dependable public and economic value.

Disclosure statement

Author confirmation required before submission. Funding, employment, advisory, financial, and nonfinancial interests must be completed and verified by Angelis Pseftis before submission. No affiliation, funding source, institutional endorsement, or conflict should be inferred from this working manuscript.

Generative artificial intelligence tools assisted with candidate-source discovery, claim extraction, bibliography reconciliation, figure-code generation, drafting, analytical structure, and editorial revision. AI systems were not treated as authors, evidence, or authoritative sources. Original sources were inspected through an AI-assisted audit, and material claim dispositions and unresolved items are recorded in the supplement. Angelis Pseftis must independently

review the cited primary sources, approve or revise every material claim, resolve disclosures, and accept responsibility for the manuscript before submission. This statement records the workflow used; it does not claim completed human verification or independent peer review.

Acknowledgments

No acknowledgments are included in this working draft. The author should add contributors, reviewers, data providers, or institutional support only after confirming their roles and obtaining any required permission.

Data, code, and supplementary materials statement

The accompanying audit package contains the claim–evidence ledger, claim-by-claim audit trail, source register, search log, reference reconciliation, benchmark-validity assessments, maturity rubric, priority tiers, scenario assumptions, figure source data, figure-generation code, and QA reports. It does not redistribute copyrighted source documents or proprietary evaluation data. No Harvard Dataverse persistent identifier has been obtained. Before formal HDSR submission, Angelis Pseftis should approve a license-compliant deposit in the repository required by the journal and add its persistent identifier.

Appendix A. Benchmark and Evidence Evaluation Template

Each consequential quantitative claim should be accompanied by the following fields.

Table A1. Benchmark and evidence evaluation fields

Field	Required content
Claim identifier	Stable identifier linked to the manuscript
Exact claim	Wording narrow enough to verify
Source	Primary publication, standard, record, or system report
Evidence grade	E0–E4
Transfer grade	T0–T4
Evaluated system	Model, version, date, and configuration
Task and dataset	Benchmark or operational task
Tools and scaffolding	Retrieval, code, search, memory, retries, verifier
Inference budget	Tokens, samples, time, or compute where available
Outcome	Numerator, denominator, metric, and uncertainty
Contamination risk	Evidence of training exposure or optimization
Operational relevance	Relationship to intended use
Contradictory evidence	Results that narrow or challenge the claim
Verification status	Source-inspection method, reviewer, date, and replication status; do not imply human verification unless completed

Appendix B. Maturity Assessment Questions

An assessment team should answer the following before assigning capability or assurance maturity:

What exact function, environment, user, and consequence are being rated?

What evidence shows that the task is performed correctly?

Has the result been reproduced independently?

How does performance vary across time, location, user, and data shift?

What are the worst observed failures and their consequences?

Does confidence predict error, and what happens when the system abstains?

What identity and authority does the system possess?

Can untrusted content change instructions or tool behavior?

Which controls are deterministic and independent of the model?

Can every consequential action be attributed and reconstructed?

How are errors detected, contained, and reversed?

What model, prompt, data, tool, or policy changes trigger re-evaluation?

What incident and near-miss evidence exists?

Who accepts the residual risk?

What evidence would invalidate the current maturity rating?

Appendix C. Minimum Reporting Standard for Agent Evaluations

Agent evaluation reports should state:

the complete system configuration, not only the base model;

task-source dates and contamination controls;

environmental reset and state-persistence rules;

available tools, permissions, credentials, and network access;

token, time, monetary, and retry budgets;

success, partial success, side effects, and prohibited actions;

performance at multiple reliability percentiles;

human intervention and review time;

monitor and verifier performance;

failure-recovery behavior;

model and tool version changes during the study; and

source code or sufficiently detailed reproduction instructions.

Appendix D. Research Questions for a Five-Year Program

How can 99 percent and 99.9 percent task horizons be estimated with practical sample sizes and realistic task diversity?

Which verifier architectures remain effective when generators can adapt to or infer the verification process?

How should authorization be delegated and revoked across thousands of ephemeral agents?

Which forms of external state minimize compounding memory error without preventing useful adaptation?

Can uncertainty estimates remain calibrated under tool use, retrieval, and distribution shift?

Which interpretability methods predict behavior under causal intervention and model updates?

How can incident data be shared across organizations while protecting security, privacy, and liability interests?

What supervisory span can humans sustain for concurrent AI agents, and what interface designs preserve situation awareness?

How should total-system energy and cost be measured per verified outcome?

Which material-change triggers provide a workable basis for continuous AI assurance?

References

Autio, C., Schwartz, R., Dunietz, J., Jain, S., Stanley, M., Tabassi, E., Hall, P., & Roberts, K. (2024). Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile (NIST AI 600-1). National Institute of Standards and Technology. https://doi.org/10.6028/NIST.AI.600-1

Bai, Y., Lv, X., Zhang, J., Lyu, H., Tang, J., Huang, Z., Du, Z., Liu, X., Zeng, A., Hou, L., Dong, Y., Tang, J., & Li, J. (2024). LongBench: A bilingual, multitask benchmark for long context understanding. In Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers) (pp. 3119–3137). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.acl-long.172

Bainbridge, L. (1983). Ironies of automation. Automatica, 19(6), 775–779. https://doi.org/10.1016/0005-1098(83)90046-8

Barrio-Hernandez, I., Yeo, J., Jänes, J., Mirdita, M., Gilchrist, C. L. M., Wein, T., Varadi, M., Velankar, S., Beltrao, P., & Steinegger, M. (2023). Clustering predicted structures at the scale of the known protein universe. Nature, 622(7983), 637–645. https://doi.org/10.1038/s41586-023-06510-w

Becker, J., Rush, N., Barnes, E., & Rein, D. (2025). Measuring the impact of early-2025 AI on experienced open-source developer productivity. arXiv. https://arxiv.org/abs/2507.09089

Becker, J., Rush, N., Cunningham, T., Rein, D., & Mahamud, K. (2026, February 24). We are changing our developer productivity experiment design. METR. https://metr.org/blog/2026-02-24-uplift-update/

Besiroglu, T., Bergerson, S. A., Michael, A., Heim, L., Luo, X., & Thompson, N. (2024). The compute divide in machine learning: A threat to academic contribution and scrutiny? arXiv. https://arxiv.org/abs/2401.02452

Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., von Arx, S., Bernstein, M. S., Bohg, J., Bosselut, A., Brunskill, E., Brynjolfsson, E., Buch, S., Card, D., Castellon, R., Chatterji, N., Chen, A., Creel, K., Davis, J. Q., Demszky, D., … Liang, P. (2021). On the opportunities and risks of foundation models. arXiv. https://arxiv.org/abs/2108.07258

Bricken, T., Templeton, A., Batson, J., Chen, B., Jermyn, A., Conerly, T., Turner, N. L., Anil, C., Denison, C., Askell, A., Lasenby, R., Wu, Y., Kravec, S., Schiefer, N., Maxwell, T., Joseph, N., Hatfield-Dodds, Z., Tamkin, A., Nguyen, K., … Olah, C. (2023). Towards monosemanticity: Decomposing language models with dictionary learning. Transformer Circuits Thread. https://transformer-circuits.pub/2023/monosemantic-features/index.html

Brohan, A., Brown, N., Carbajal, J., Chebotar, Y., Chen, X., Choromanski, K. M., Ding, T., Driess, D., Dubey, K. A., Finn, C., Florence, P., Fu, C., Gonzalez Arenas, M., Gopalakrishnan, K., Han, K., Hausman, K., Herzog, A., Hsu, J., Ichter, B., … Zitkovich, B. (2023). RT-2: Vision-language-action models transfer web knowledge to robotic control. In Proceedings of the 7th Conference on Robot Learning (PMLR Vol. 229, pp. 2165–2183). https://proceedings.mlr.press/v229/zitkovich23a.html

Casper, S., Davies, X., Shi, C., Gilbert, T. K., Scheurer, J., Rando, J., Freedman, R., Korbak, T., Lindner, D., Freire, P., Wang, T., Marks, S., Segerie, C.-R., Carroll, M., Peng, A., Christoffersen, P., Damani, M., Slocum, S., Anwar, U., … Hadfield-Menell, D. (2023). Open problems and fundamental limitations of reinforcement learning from human feedback. Transactions on Machine Learning Research. https://openreview.net/forum?id=bx24KpJ4Eb

Driess, D., Xia, F., Sajjadi, M. S. M., Lynch, C., Chowdhery, A., Ichter, B., Wahid, A., Tompson, J., Vuong, Q., Yu, T., Huang, W., Chebotar, Y., Sermanet, P., Duckworth, D., Levine, S., Vanhoucke, V., Hausman, K., Toussaint, M., Greff, K., … Florence, P. (2023). PaLM-E: An embodied multimodal language model. In Proceedings of the 40th International Conference on Machine Learning (PMLR Vol. 202, pp. 8469–8488). https://proceedings.mlr.press/v202/driess23a.html

Dziemian, M., Lin, M., Fu, X., Nowak, M., Winter, N., Jones, E., Zou, A., Ahmad, L., Chaudhuri, K., Chennabasappa, S., Davies, X., Deason, L., Edelman, B. L., Emek, T., Evtimov, I., Gust, J., Hamin, M., He, K., Krawiecka, K., … Kolter, J. Z. (2026). How vulnerable are AI agents to indirect prompt injections? Insights from a large-scale public competition. arXiv. https://arxiv.org/abs/2603.15714

Elhage, N., Nanda, N., Olsson, C., Henighan, T., Joseph, N., Mann, B., Askell, A., Bai, Y., Chen, A., Conerly, T., DasSarma, N., Drain, D., Ganguli, D., Hatfield-Dodds, Z., Hernandez, D., Jones, A., Kernion, J., Lovitt, L., Ndousse, K., … Olah, C. (2021). A mathematical framework for transformer circuits. Transformer Circuits Thread. https://transformer-circuits.pub/2021/framework/index.html

European Parliament & Council of the European Union. (2024). Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence. Official Journal of the European Union, L 2024/1689. https://eur-lex.europa.eu/eli/reg/2024/1689/oj

European Parliament & Council of the European Union. (2026). Regulation (EU) 2026/1744 amending Regulations (EU) 2024/1689, (EU) 2018/1139 and (EU) 2023/1230 as regards the simplification of the implementation of harmonised rules on artificial intelligence. Official Journal of the European Union, L 2026/1744. http://data.europa.eu/eli/reg/2026/1744/oj

European Union. (2026). Consolidated text: Regulation (EU) 2024/1689 of the European Parliament and of the Council laying down harmonised rules on artificial intelligence (current to 27 July 2026). EUR-Lex. https://eur-lex.europa.eu/eli/reg/2024/1689/2026-07-27/eng/pdf

Farquhar, S., Kossen, J., Kuhn, L., & Gal, Y. (2024). Detecting hallucinations in large language models using semantic entropy. Nature, 630(8017), 625–630. https://doi.org/10.1038/s41586-024-07421-0

Fawzi, A., Balog, M., Huang, A., Hubert, T., Romera-Paredes, B., Barekatain, M., Novikov, A., Ruiz, F. J. R., Schrittwieser, J., Swirszcz, G., Silver, D., Hassabis, D., & Kohli, P. (2022). Discovering faster matrix multiplication algorithms with reinforcement learning. Nature, 610(7930), 47–53. https://doi.org/10.1038/s41586-022-05172-4

Fu, C., Dai, Y., Luo, Y., Li, L., Ren, S., Zhang, R., Wang, Z., Zhou, C., Shen, Y., Zhang, M., Chen, P., Li, Y., Lin, S., Zhao, S., Li, K., Xu, T., Zheng, X., Chen, E., Shan, C., … Sun, X. (2025). Video-MME: The first-ever comprehensive evaluation benchmark of multi-modal LLMs in video analysis. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 24108–24118). https://doi.org/10.1109/CVPR52734.2025.02245

Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J. W., Wallach, H., Daumé III, H., & Crawford, K. (2021). Datasheets for datasets. Communications of the ACM, 64(12), 86–92. https://doi.org/10.1145/3458723

Geifman, Y., & El-Yaniv, R. (2019). SelectiveNet: A deep neural network with an integrated reject option. In Proceedings of the 36th International Conference on Machine Learning (PMLR Vol. 97, pp. 2151–2159). https://proceedings.mlr.press/v97/geifman19a.html

Geirhos, R., Jacobsen, J.-H., Michaelis, C., Zemel, R., Brendel, W., Bethge, M., & Wichmann, F. A. (2020). Shortcut learning in deep neural networks. Nature Machine Intelligence, 2(11), 665–673. https://doi.org/10.1038/s42256-020-00257-z

Gneiting, T., & Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. Journal of the American Statistical Association, 102(477), 359–378. https://doi.org/10.1198/016214506000001437

Greshake, K., Abdelnabi, S., Mishra, S., Endres, C., Holz, T., & Fritz, M. (2023). Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection. In Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security (pp. 79–90). https://doi.org/10.1145/3605764.3623985

Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On calibration of modern neural networks. In Proceedings of the 34th International Conference on Machine Learning (PMLR Vol. 70, pp. 1321–1330). https://proceedings.mlr.press/v70/guo17a.html

Hafner, D., Pasukonis, J., Ba, J., & Lillicrap, T. (2025). Mastering diverse control tasks through world models. Nature, 640(8059), 647–653. https://doi.org/10.1038/s41586-025-08744-2

Hanley, J. A., & Lippman-Hand, A. (1983). If nothing goes wrong, is everything all right? Interpreting zero numerators. JAMA, 249(13), 1743–1745. https://doi.org/10.1001/jama.1983.03330370053031

Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., & Steinhardt, J. (2021). Measuring massive multitask language understanding. International Conference on Learning Representations. https://openreview.net/forum?id=d7KBjmI3GmQ

Hendrycks, D., Zhao, K., Basart, S., Steinhardt, J., & Song, D. (2021). Natural adversarial examples. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 15262–15271). https://doi.org/10.1109/CVPR46437.2021.01501

Hoffmann, J., Borgeaud, S., Mensch, A., Buchatskaya, E., Cai, T., Rutherford, E., de Las Casas, D., Hendricks, L. A., Welbl, J., Clark, A., Hennigan, T., Noland, E., Millican, K., van den Driessche, G., Damoc, B., Guy, A., Osindero, S., Simonyan, K., Elsen, E., … Sifre, L. (2022). Training compute-optimal large language models. Advances in Neural Information Processing Systems, 35, 30016–30030. https://doi.org/10.52202/068431-2176

Hsieh, C.-P., Sun, S., Kriman, S., Acharya, S., Rekesh, D., Jia, F., Zhang, Y., & Ginsburg, B. (2024). RULER: What's the real context size of your long-context language models? Conference on Language Modeling. https://openreview.net/forum?id=kIoBbc76Sy

Hubinger, E., Denison, C., Mu, J., Lambert, M., Tong, M., MacDiarmid, M., Lanham, T., Ziegler, D. M., Maxwell, T., Cheng, N., Jermyn, A., Askell, A., Radhakrishnan, A., Anil, C., Duvenaud, D., Ganguli, D., Barez, F., Clark, J., Ndousse, K., … Perez, E. (2024). Sleeper agents: Training deceptive LLMs that persist through safety training. arXiv. https://arxiv.org/abs/2401.05566

International Organization for Standardization, International Electrotechnical Commission, & Institute of Electrical and Electronics Engineers. (2022). ISO/IEC/IEEE 15026-2:2022: Systems and software engineering—Systems and software assurance—Part 2: Assurance case. https://www.iso.org/standard/80625.html

International Organization for Standardization. (2023a). ISO/IEC 23894:2023: Information technology—Artificial intelligence—Guidance on risk management. https://www.iso.org/standard/77304.html

International Organization for Standardization. (2023b). ISO/IEC 42001:2023: Information technology—Artificial intelligence—Management system. https://www.iso.org/standard/42001

Jimenez, C. E., Yang, J., Wettig, A., Yao, S., Pei, K., Press, O., & Narasimhan, K. (2024). SWE-bench: Can language models resolve real-world GitHub issues? International Conference on Learning Representations. https://openreview.net/forum?id=VTF8yNQM66

Jin, Z., Chen, Y., Leeb, F., Gresele, L., Kamal, O., Lyu, Z., Blin, K., Gonzalez Adauto, F., Kleiman-Weiner, M., Sachan, M., & Schölkopf, B. (2023). CLadder: Assessing causal reasoning in language models. Advances in Neural Information Processing Systems, 36, 31038–31065. https://doi.org/10.52202/075280-1353

Jumper, J., Evans, R., Pritzel, A., Green, T., Figurnov, M., Ronneberger, O., Tunyasuvunakool, K., Bates, R., Žídek, A., Potapenko, A., Bridgland, A., Meyer, C., Kohl, S. A. A., Ballard, A. J., Cowie, A., Romera-Paredes, B., Nikolov, S., Jain, R., Adler, J., … Hassabis, D. (2021). Highly accurate protein structure prediction with AlphaFold. Nature, 596(7873), 583–589. https://doi.org/10.1038/s41586-021-03819-2

Kadavath, S., Conerly, T., Askell, A., Henighan, T., Drain, D., Perez, E., Schiefer, N., Hatfield-Dodds, Z., DasSarma, N., Tran-Johnson, E., Johnston, S., El-Showk, S., Jones, A., Elhage, N., Hume, T., Chen, A., Bai, Y., Bowman, S., Fort, S., … Kaplan, J. (2022). Language models (mostly) know what they know. arXiv. https://arxiv.org/abs/2207.05221

Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., Gray, S., Radford, A., Wu, J., & Amodei, D. (2020). Scaling laws for neural language models. arXiv. https://arxiv.org/abs/2001.08361

Kiela, D., Bartolo, M., Nie, Y., Kaushik, D., Geiger, A., Wu, Z., Vidgen, B., Prasad, G., Singh, A., Ringshia, P., Ma, Z., Thrush, T., Riedel, S., Waseem, Z., Stenetorp, P., Jia, R., Bansal, M., Potts, C., & Williams, A. (2021). Dynabench: Rethinking benchmarking in NLP. In Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (pp. 4110–4124). Association for Computational Linguistics. https://doi.org/10.18653/v1/2021.naacl-main.324

Kinniment, M., Sato, L. J. K., Du, H., Goodrich, B., Hasin, M., Chan, L., Miles, L. H., Lin, T. R., Wijk, H., Burget, J., Ho, A., Barnes, E., & Christiano, P. (2023). Evaluating language-model agents on realistic autonomous tasks. arXiv. https://arxiv.org/abs/2312.11671

Koh, P. W., Sagawa, S., Marklund, H., Xie, S. M., Zhang, M., Balsubramani, A., Hu, W., Yasunaga, M., Phillips, R. L., Gao, I., Lee, T., David, E., Stavness, I., Guo, W., Earnshaw, B., Haque, I., Beery, S. M., Leskovec, J., Kundaje, A., … Liang, P. (2021). WILDS: A benchmark of in-the-wild distribution shifts. In Proceedings of the 38th International Conference on Machine Learning (PMLR Vol. 139, pp. 5637–5664). https://proceedings.mlr.press/v139/koh21a.html

Krakovna, V., Uesato, J., Mikulik, V., Rahtz, M., Everitt, T., Kumar, R., Kenton, Z., Leike, J., & Legg, S. (2020, April 21). Specification gaming: The flip side of AI ingenuity. Google DeepMind. https://deepmind.google/blog/specification-gaming-the-flip-side-of-ai-ingenuity/

Kusano, K. D., Scanlon, J. M., Chen, Y.-H., McMurry, T. L., Chen, R., Gode, T., & Victor, T. (2024). Comparison of Waymo rider-only crash data to human benchmarks at 7.1 million miles. Traffic Injury Prevention, 25(Suppl. 1), S66–S77. https://doi.org/10.1080/15389588.2024.2380786

Kwa, T., West, B., Becker, J., Deng, A., Garcia, K., Hasin, M., Jawhar, S., Kinniment, M., Rush, N., Von Arx, S., Bloom, R., Broadley, T., Du, H., Goodrich, B., Jurkovic, N., Miles, L., Nix, S., Lin, T., Parikh, N., … Chan, L. (2025). Measuring AI ability to complete long software tasks. Advances in Neural Information Processing Systems, 38, 102382–102435. https://doi.org/10.52202/085713-3086

Kıcıman, E., Ness, R., Sharma, A., & Tan, C. (2023). Causal reasoning and large language models: Opening a new frontier for causality. arXiv. https://arxiv.org/abs/2305.00050

Langosco, L. L. D., Koch, J., Sharkey, L. D., Pfau, J., & Krueger, D. (2022). Goal misgeneralization in deep reinforcement learning. In Proceedings of the 39th International Conference on Machine Learning (PMLR Vol. 162, pp. 12004–12019). https://proceedings.mlr.press/v162/langosco22a.html

Li, Y., Choi, D., Chung, J., Kushman, N., Schrittwieser, J., Leblond, R., Eccles, T., Keeling, J., Gimeno, F., Dal Lago, A., Hubert, T., Choy, P., de Masson d'Autume, C., Babuschkin, I., Chen, X., Huang, P.-S., Welbl, J., Gowal, S., Cherepanov, A., … Vinyals, O. (2022). Competition-level code generation with AlphaCode. Science, 378(6624), 1092–1097. https://doi.org/10.1126/science.abq1158

Liang, P., Bommasani, R., Lee, T., Tsipras, D., Soylu, D., Yasunaga, M., Zhang, Y., Narayanan, D., Wu, Y., Kumar, A., Newman, B., Yuan, B., Yan, B., Zhang, C., Cosgrove, C., Manning, C. D., Ré, C., Acosta-Navas, D., Hudson, D. A., … Koreeda, Y. (2023). Holistic evaluation of language models. Transactions on Machine Learning Research. https://openreview.net/forum?id=iO4LZibEqW

Lin, S., Hilton, J., & Evans, O. (2022). TruthfulQA: Measuring how models mimic human falsehoods. In Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers) (pp. 3214–3252). Association for Computational Linguistics. https://doi.org/10.18653/v1/2022.acl-long.229

Liu, N. F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., & Liang, P. (2024). Lost in the middle: How language models use long contexts. Transactions of the Association for Computational Linguistics, 12, 157–173. https://doi.org/10.1162/tacl_a_00638

Liu, X., Yu, H., Zhang, H., Xu, Y., Lei, X., Lai, H., Gu, Y., Ding, H., Men, K., Yang, K., Zhang, S., Deng, X., Zeng, A., Du, Z., Zhang, C., Shen, S., Zhang, T., Su, Y., Sun, H., … Tang, J. (2024). AgentBench: Evaluating LLMs as agents. International Conference on Learning Representations. https://openreview.net/forum?id=zAdUB0aCTQ

Manakul, P., Liusie, A., & Gales, M. J. F. (2023). SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (pp. 9004–9017). Association for Computational Linguistics. https://doi.org/10.18653/v1/2023.emnlp-main.557

Mata v. Avianca, Inc., 678 F. Supp. 3d 443 (S.D.N.Y. 2023).

Merchant, A., Batzner, S., Schoenholz, S. S., Aykol, M., Cheon, G., & Cubuk, E. D. (2023). Scaling deep learning for materials discovery. Nature, 624(7990), 80–85. https://doi.org/10.1038/s41586-023-06735-9

METR. (2026a, June 26). Summary of METR's predeployment evaluation of GPT-5.6 Sol. https://metr.org/blog/2026-06-26-gpt-5-6-sol/

METR. (2026b, January 29). Time Horizon 1.1. https://metr.org/blog/2026-1-29-time-horizon-1-1/

Min, S., Krishna, K., Lyu, X., Lewis, M., Yih, W.-t., Koh, P., Iyyer, M., Zettlemoyer, L., & Hajishirzi, H. (2023). FActScore: Fine-grained atomic evaluation of factual precision in long-form text generation. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (pp. 12076–12100). Association for Computational Linguistics. https://doi.org/10.18653/v1/2023.emnlp-main.741

Mitchell, M., Wu, S., Zaldivar, A., Barnes, P., Vasserman, L., Hutchinson, B., Spitzer, E., Raji, I. D., & Gebru, T. (2019). Model cards for model reporting. In Proceedings of the Conference on Fairness, Accountability, and Transparency (pp. 220–229). Association for Computing Machinery. https://doi.org/10.1145/3287560.3287596

Muennighoff, N., Rush, A. M., Barak, B., Le Scao, T., Piktus, A., Tazi, N., Pyysalo, S., Wolf, T., & Raffel, C. A. (2023). Scaling data-constrained language models. Advances in Neural Information Processing Systems, 36, 50358–50376. https://doi.org/10.52202/075280-2191

National Cybersecurity Center of Excellence. (2026, February 5). Accelerating the adoption of software and AI agent identity and authorization [Draft concept paper]. National Institute of Standards and Technology. https://www.nccoe.nist.gov/publications/other/accelerating-adoption-software-and-ai-agent-identity-and-authorization-concept

National Highway Traffic Safety Administration. (2024, September 26). Consent order: Cruise; Standing General Order reporting. U.S. Department of Transportation. https://www.nhtsa.gov/document/consent-order-cruise-standing-general-order-reporting

National Institute of Standards and Technology. (2026a, August 14). AI Agent Standards Initiative. https://www.nist.gov/artificial-intelligence/ai-agent-standards-initiative

National Institute of Standards and Technology. (2026c, March 23). Insights into AI agent security from a large-scale red-teaming competition. https://www.nist.gov/blogs/caisi-research-blog/insights-ai-agent-security-large-scale-red-teaming-competition

National Institute of Standards and Technology. (2026d, January). Practices for automated benchmark evaluations of language models (Initial public draft; NIST AI 800-2 ipd). https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.800-2.ipd.pdf

Needham, J., Edkins, G., Pimpale, G., Bartsch, H., & Hobbhahn, M. (2025). Large language models often know when they are being evaluated. arXiv. https://arxiv.org/abs/2505.23836

Olsson, C., Elhage, N., Nanda, N., Joseph, N., DasSarma, N., Henighan, T., Mann, B., Askell, A., Bai, Y., Chen, A., Conerly, T., Drain, D., Ganguli, D., Hatfield-Dodds, Z., Hernandez, D., Johnston, S., Jones, A., Kernion, J., Lovitt, L., … Olah, C. (2022). In-context learning and induction heads. Transformer Circuits Thread. https://transformer-circuits.pub/2022/in-context-learning-and-induction-heads/index.html

OpenAI. (2026a, July 8). Separating signal from noise in coding evaluations. https://openai.com/index/separating-signal-from-noise-coding-evaluations/

OpenAI. (2026b, February 23). Why SWE-bench Verified no longer measures frontier coding capabilities. https://openai.com/index/why-we-no-longer-evaluate-swe-bench-verified/

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C. L., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., Schulman, J., Hilton, J., Kelton, F., Miller, L., Simens, M., Askell, A., Welinder, P., Christiano, P., Leike, J., & Lowe, R. (2022). Training language models to follow instructions with human feedback. Advances in Neural Information Processing Systems, 35, 27730–27744. https://proceedings.neurips.cc/paper_files/paper/2022/hash/b1efde53be364a73914f58805a001731-Abstract-Conference.html

O'Neill, A., Rehman, A., Maddukuri, A., Gupta, A., Padalkar, A., Lee, A., Pooley, A., Gupta, A., Mandlekar, A., Jain, A., Tung, A., Bewley, A., Herzog, A., Irpan, A., Khazatsky, A., Rai, A., Gupta, A., Wang, A., Singh, A., … Lin, Z. (2024). Open X-Embodiment: Robotic learning datasets and RT-X models. In 2024 IEEE International Conference on Robotics and Automation (pp. 6892–6903). https://doi.org/10.1109/ICRA57147.2024.10611477

Parasuraman, R., & Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. Human Factors, 39(2), 230–253. https://doi.org/10.1518/001872097778543886

Pearce, H., Ahmad, B., Tan, B., Dolan-Gavitt, B., & Karri, R. (2022). Asleep at the keyboard? Assessing the security of GitHub Copilot's code contributions. In 2022 IEEE Symposium on Security and Privacy (pp. 754–768). https://doi.org/10.1109/SP46214.2022.9833571

Peng, S., Kalliamvakou, E., Cihon, P., & Demirer, M. (2023). The impact of AI on developer productivity: Evidence from GitHub Copilot. arXiv. https://arxiv.org/abs/2302.06590

Perry, N., Srivastava, M., Kumar, D., & Boneh, D. (2023). Do users write more insecure code with AI assistants? In Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security (pp. 2785–2799). Association for Computing Machinery. https://doi.org/10.1145/3576915.3623157

Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., & Sutskever, I. (2023). Robust speech recognition via large-scale weak supervision. In Proceedings of the 40th International Conference on Machine Learning (PMLR Vol. 202, pp. 28492–28518). https://proceedings.mlr.press/v202/radford23a.html

Rein, D., Hou, B. L., Stickland, A. C., Petty, J., Pang, R. Y., Dirani, J., Michael, J., & Bowman, S. R. (2024). GPQA: A graduate-level Google-proof Q&A benchmark. Conference on Language Modeling. https://openreview.net/forum?id=Ti67584b98

Riggs, J., Hamin, M., Perry, N., Edelman, B., & Cihon, P. (2026, May 18). Summary analysis of responses to the Request for Information regarding security considerations for AI agents (NIST AI 800-5). National Institute of Standards and Technology. https://www.nist.gov/publications/summary-analysis-responses-request-information-regarding-security-considerations-ai

Romera-Paredes, B., Barekatain, M., Novikov, A., Balog, M., Kumar, M. P., Dupont, E., Ruiz, F. J. R., Ellenberg, J. S., Wang, P., Fawzi, O., Kohli, P., & Fawzi, A. (2024). Mathematical discoveries from program search with large language models. Nature, 625(7995), 468–475. https://doi.org/10.1038/s41586-023-06924-6

Rose, S., Borchert, O., Mitchell, S., & Connelly, S. (2020). Zero trust architecture (NIST SP 800-207). National Institute of Standards and Technology. https://doi.org/10.6028/NIST.SP.800-207

Ross, R., Winstead, M., & McEvilley, M. (2022). Engineering trustworthy secure systems (NIST SP 800-160 Vol. 1 Rev. 1). National Institute of Standards and Technology. https://doi.org/10.6028/NIST.SP.800-160v1r1

Saltzer, J. H., & Schroeder, M. D. (1975). The protection of information in computer systems. Proceedings of the IEEE, 63(9), 1278–1308. https://doi.org/10.1109/PROC.1975.9939

Sardana, N., Portes, J., Doubov, S., & Frankle, J. (2024). Beyond Chinchilla-optimal: Accounting for inference in language model scaling laws. In Proceedings of the 41st International Conference on Machine Learning (PMLR Vol. 235, pp. 43445–43460). https://proceedings.mlr.press/v235/sardana24a.html

Schaeffer, R., Miranda, B., & Koyejo, S. (2023). Are emergent abilities of large language models a mirage? Advances in Neural Information Processing Systems, 36, 55565–55581. https://doi.org/10.52202/075280-2425

Schick, T., Dwivedi-Yu, J., Dessì, R., Raileanu, R., Lomeli, M., Hambro, E., Zettlemoyer, L., Cancedda, N., & Scialom, T. (2023). Toolformer: Language models can teach themselves to use tools. Advances in Neural Information Processing Systems, 36, 68539–68551. https://doi.org/10.52202/075280-2997

Schrittwieser, J., Antonoglou, I., Hubert, T., Simonyan, K., Sifre, L., Schmitt, S., Guez, A., Lockhart, E., Hassabis, D., Graepel, T., Lillicrap, T., & Silver, D. (2020). Mastering Atari, Go, chess and shogi by planning with a learned model. Nature, 588(7839), 604–609. https://doi.org/10.1038/s41586-020-03051-4

Sculley, D., Holt, G., Golovin, D., Davydov, E., Phillips, T., Ebner, D., Chaudhary, V., Young, M., Crespo, J.-F., & Dennison, D. (2015). Hidden technical debt in machine learning systems. Advances in Neural Information Processing Systems, 28, 2503–2511. https://proceedings.neurips.cc/paper_files/paper/2015/hash/86df7dcfd896fcaf2674f757a2463eba-Abstract.html

Seamless Communication, Barrault, L., Chung, Y.-A., Meglioli, M. C., Dale, D., Dong, N., Duquenne, P.-A., Elsahar, H., Gong, H., Heffernan, K., Hoffman, J., Klaiber, C., Li, P., Licht, D., Maillard, J., Rakotoarison, A., Sadagopan, K. R., Wenzek, G., Ye, E., … Wang, S. (2023). SeamlessM4T: Massively multilingual & multimodal machine translation. arXiv. https://arxiv.org/abs/2308.11596

Shehabi, A., Smith, S. J., Hubbard, A., Newkirk, A., Lei, N., Siddik, M. A. B., Holecek, B., Koomey, J. G., Masanet, E. R., & Sartor, D. A. (2024). 2024 United States Data Center Energy Usage Report (LBNL-2001637). Lawrence Berkeley National Laboratory. https://doi.org/10.71468/P1WC7Q

Shinn, N., Cassano, F., Gopinath, A., Narasimhan, K., & Yao, S. (2023). Reflexion: Language agents with verbal reinforcement learning. Advances in Neural Information Processing Systems, 36, 8634–8652. https://doi.org/10.52202/075280-0377

Shumailov, I., Shumaylov, Z., Zhao, Y., Papernot, N., Anderson, R., & Gal, Y. (2024). AI models collapse when trained on recursively generated data. Nature, 631(8022), 755–759. https://doi.org/10.1038/s41586-024-07566-y

Smith, S. J., Hubbard, A., Newkirk, A., Ganeshalingam, M., Holecek, B., Sartor, D. A., Mills, M., & Shehabi, A. (2026, June). United States Data Center Energy Usage Report: 2025 Update. Lawrence Berkeley National Laboratory. https://eta-publications.lbl.gov/publications/united-states-data-center-energy-2025

Srivastava, A., Rastogi, A., Rao, A., Shoeb, A. A. M., Abid, A., Fisch, A., Brown, A. R., Santoro, A., Gupta, A., Garriga-Alonso, A., Kluska, A., Lewkowycz, A., Agarwal, A., Power, A., Ray, A., Warstadt, A., Kocurek, A. W., Safaya, A., Tazarv, A., … Wu, Z. (2023). Beyond the imitation game: Quantifying and extrapolating the capabilities of language models. Transactions on Machine Learning Research. https://openreview.net/forum?id=uyTL5Bvosj

Tabassi, E. (2023). Artificial intelligence risk management framework (AI RMF 1.0) (NIST AI 100-1). National Institute of Standards and Technology. https://doi.org/10.6028/NIST.AI.100-1

Templeton, A., Conerly, T., Marcus, J., Lindsey, J., Bricken, T., Chen, B., Pearce, A., Citro, C., Ameisen, E., Jones, A., Cunningham, H., Turner, N. L., McDougall, C., MacDiarmid, M., Tamkin, A., Durmus, E., Hume, T., Mosconi, F., Freeman, C. D., … Henighan, T. (2024). Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet. Transformer Circuits Thread. https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html

Teoh, E. R., Kidd, D. G., & Riexinger, L. E. (2026, July). Rise of the machines: Crash experiences of highly automated vehicles and human drivers. Insurance Institute for Highway Safety. https://www.iihs.org/research-areas/bibliography/ref/2374

Trinh, T. H., Wu, Y., Le, Q. V., He, H., & Luong, T. (2024). Solving olympiad geometry without human demonstrations. Nature, 625(7995), 476–482. https://doi.org/10.1038/s41586-023-06747-5

U.S. Attorney's Office, Northern District of California. (2024, November 14). Cruise admits to submitting a false report to influence a federal investigation and agrees to pay $500,000. U.S. Department of Justice. https://www.justice.gov/usao-ndca/pr/cruise-admits-submitting-false-report-influence-federal-investigation-and-agrees-pay

U.S. Government Accountability Office. (2025a, April 22). Artificial intelligence: Generative AI's environmental and human effects (GAO-25-107172). https://www.gao.gov/products/gao-25-107172

U.S. Government Accountability Office. (2025b, December 11). Semiconductors: Information on projects funded to strengthen U.S. supply chain (GAO-26-107882). https://www.gao.gov/products/gao-26-107882

Varadi, M., Anyango, S., Deshpande, M., Nair, S., Natassia, C., Yordanova, G., Yuan, D., Stroe, O., Wood, G., Laydon, A., Žídek, A., Green, T., Tunyasuvunakool, K., Petersen, S., Jumper, J., Clancy, E., Green, R., Vora, A., Lutfi, M., … Velankar, S. (2022). AlphaFold Protein Structure Database: Massively expanding the structural coverage of protein-sequence space with high-accuracy models. Nucleic Acids Research, 50(D1), D439–D444. https://doi.org/10.1093/nar/gkab1061

Vassilev, A., Oprea, A., Fordyce, A., Anderson, H., Davies, X., & Hamin, M. (2025). Adversarial machine learning: A taxonomy and terminology of attacks and mitigations (NIST AI 100-2e2025). National Institute of Standards and Technology. https://doi.org/10.6028/NIST.AI.100-2e2025

Wang, Y., Ma, X., Zhang, G., Ni, Y., Chandra, A., Guo, S., Ren, W., Arulraj, A., He, X., Jiang, Z., Li, T., Ku, M., Wang, K., Zhuang, A., Fan, R., Yue, X., & Chen, W. (2024). MMLU-Pro: A more robust and challenging multi-task language understanding benchmark. Advances in Neural Information Processing Systems, 37, 95266–95290. https://doi.org/10.52202/079017-3018

White, C., Dooley, S., Roberts, M., Pal, A., Feuer, B., Jain, S., Shwartz-Ziv, R., Jain, N., Saifullah, K., Dey, S., Shubh-Agrawal, Sandha, S. S., Naidu, S., Hegde, C., LeCun, Y., Goldstein, T., Neiswanger, W., & Goldblum, M. (2025). LiveBench: A challenging, contamination-limited LLM benchmark. International Conference on Learning Representations. https://proceedings.iclr.cc/paper_files/paper/2025/hash/e4a46394ba5378b3f9a186a5b4c650d1-Abstract-Conference.html

Xie, T., Zhang, D., Chen, J., Li, X., Zhao, S., Cao, R., Hua, T. J., Cheng, Z., Shin, D., Lei, F., Liu, Y., Xu, Y., Zhou, S., Savarese, S., Xiong, C., Zhong, V., & Yu, T. (2024). OSWorld: Benchmarking multimodal agents for open-ended tasks in real computer environments. Advances in Neural Information Processing Systems, 37, 52040–52094. https://doi.org/10.52202/079017-1650

Yang, S., Chiang, W.-L., Zheng, L., Gonzalez, J. E., & Stoica, I. (2024). Rethinking benchmark and contamination for language models with rephrased samples. arXiv. https://arxiv.org/abs/2311.04850

Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., & Cao, Y. (2023). ReAct: Synergizing reasoning and acting in language models. International Conference on Learning Representations. https://openreview.net/forum?id=WE_vluYUL-X

Yu, W., Yang, Z., Li, L., Wang, J., Lin, K., Liu, Z., Wang, X., & Wang, L. (2024). MM-Vet: Evaluating large multimodal models for integrated capabilities. In Proceedings of the 41st International Conference on Machine Learning (PMLR Vol. 235, pp. 57730–57754). https://proceedings.mlr.press/v235/yu24o.html

Yuan, M., Zhou, Z., Xiong, X., Wu, W., Sun, J., Song, J., Cui, K., Wang, B., Wu, H., Li, Y., Lu, D., Lu, H., Zhen, Q., Wang, X., Deng, J., Yang, Y., Chen, C., Zheng, B., Su, A., … Yu, T. (2026). OSWorld 2.0: Benchmarking computer use agents on long-horizon real-world tasks. arXiv. https://arxiv.org/abs/2606.29537

Yue, X., Ni, Y., Zheng, T., Zhang, K., Liu, R., Zhang, G., Stevens, S., Jiang, D., Ren, W., Sun, Y., Wei, C., Yu, B., Yuan, R., Sun, R., Yin, M., Zheng, B., Yang, Z., Liu, Y., Huang, W., … Chen, W. (2024). MMMU: A massive multi-discipline multimodal understanding and reasoning benchmark for expert AGI. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 9556–9567). https://doi.org/10.1109/CVPR52733.2024.00913

Zhang, L., He, S., Zhang, C., Kang, Y., Li, B., Xie, C., Wang, J., Wang, M., Huang, Y., Fu, S., Nallipogu, E., Lin, Q., Dang, Y., Rajmohan, S., & Zhang, D. (2025). SWE-bench Goes Live! arXiv. https://arxiv.org/abs/2505.23419

Zhou, S., Xu, F. F., Zhu, H., Zhou, X., Lo, R., Sridhar, A., Cheng, X., Ou, T., Bisk, Y., Fried, D., Alon, U., & Neubig, G. (2024). WebArena: A realistic web environment for building autonomous agents. International Conference on Learning Representations. https://openreview.net/forum?id=rmiwIL98uQ

Zou, A., Wang, Z., Carlini, N., Nasr, M., Kolter, J. Z., & Fredrikson, M. (2023). Universal and transferable adversarial attacks on aligned language models. arXiv. https://arxiv.org/abs/2307.15043